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Dissociable after-effects of prosocial acts: Effort is costly for others but valued for self

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This paper provides **solid** electrophysiological evidence that an individual's effort expenditure increases the subjective value of a subsequent reward when the beneficiary is the individual themselves, but decreases the subjective value of a reward when the beneficiary is someone else. These findings have **valuable** implications for our understanding of how effort investment shapes reward evaluation during prosocial behavior.

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Abstract

Engaging in prosocial behavior requires effort, yet people are often averse to exerting effort for others' benefit. However, it remains unclear how effort exertion affects subsequent reward evaluation during prosocial acts. Here, we combined high-temporal-resolution electroencephalography with a paradigm that independently manipulated effort and reward for self and others to elucidate the neural mechanisms underlying the reward after-effect of prosocial effort expenditure. We found dissociable reward after-effects for self-benefiting and other-benefiting effort. For self-benefiting rewards, the reward positivity (RewP) increased with effort demand, suggesting an effort-enhancement effect. In contrast, for other-benefiting rewards, the RewP decreased as effort increased, demonstrating an effort-discounting effect. Critically, this dissociation was contingent upon high reward magnitude and modulated by individual differences in effort discounting, yet remained distinct from performance evaluation. Our findings reveal distinct neural computations for self- and other-benefiting efforts, offering new insights into how prior effort expenditure shapes reward evaluation during prosocial behavior.

1 Introduction

Prosocial behaviors are acts that benefit others and often involve some personal costs (Bierhoff, 2002 [↗](#)). These actions are crucial for promoting individual physical (Post, 2005 [↗](#)) and mental (Raposa et al., 2016 [↗](#)) well-being and serve as a powerful force enhancing social cohesiveness and group bonding (Fehr & Fischbacher, 2003 [↗](#)). According to the cost-benefit framework (Contreras-Huerta et al., 2020 [↗](#)), people help others either because they are highly sensitive to others' welfare or because they are less sensitive to their own costs. However, prior studies have predominantly manipulated only financial costs (Engel, 2011 [↗](#)), ignoring the most common cost type: effort, a non-contextual factor that represents the physical or cognitive resources required for goal-directed behavior (Shenhav et al., 2017 [↗](#)). Indeed, prosocial behavior in everyday life

requires the investment of varying amounts of effort, whether helping a colleague proofread a paper or holding an elevator for a stranger. In this study, we examined how effort exertion affects subsequent reward evaluation during prosocial acts.

Effort is typically considered costly and aversive. All else being equal, people usually follow a “law of less work” (Hull, 1943 [↗](#)) and prefer lower over higher task demands (Kool et al., 2010 [↗](#)), manifesting as effort discounting (Westbrook et al., 2013 [↗](#)). Due to its inherent aversiveness, effort serves as an ideal proxy for cost in studying prosocial behavior. Recent studies have characterized the psychological and neurobiological mechanisms underlying prosocial effort. In these studies, participants repeatedly decide whether to invest effort to gain financial rewards for themselves or others. A consistent finding is that participants are less willing to exert effort to benefit others than themselves, specifically when the required effort is high. This prosocial apathy is observed for both physical effort (Lockwood et al., 2017 [↗](#)) and cognitive effort (Depow et al., 2022 [↗](#)). It is modulated by acute stress (Forbes et al., 2024 [↗](#)), aging (Lockwood et al., 2021 [↗](#)), and individual differences in broad affective traits (Contreras-Huerta et al., 2022 [↗](#)). Moreover, prosocial acts are invigorated to a lesser degree during effort exertion than identical self-benefiting ones (Lockwood et al., 2017 [↗](#); Lockwood et al., 2022 [↗](#)). Neuroimaging and brain lesion studies have revealed that prosocial effort costs are tracked by neural activity in the anterior cingulate gyrus, anterior insula, and ventromedial prefrontal cortex (Forbes et al., 2024 [↗](#); Lockwood et al., 2024 [↗](#); Lockwood et al., 2022 [↗](#)). While these studies have focused on how people choose to exert effort during decision making and energize their actions during effort exertion, they have largely ignored another important aspect of effort expenditure: the reward after-effect of effort expenditure.

The reward after-effect of effort expenditure refers to a temporary tuning toward or amplification of reward-related motivation following effort exertion (Inzlicht et al., 2018 [↗](#); Kelley et al., 2019 [↗](#)). Specifically, people assign more value to rewards they have put effort into and are more reluctant to share monetary rewards earned through greater effort (Arkes et al., 1994 [↗](#); Norton et al., 2012 [↗](#)). The effort-adds-value phenomenon is observed across species (Lydall et al., 2010 [↗](#); Pompilio et al., 2006 [↗](#)), suggesting that it is biologically hard-wired. Echoing these behavioral findings, neuroimaging evidence has demonstrated that prior effort investment increases brain activity in reward-related neural circuits including the anterior cingulate cortex, orbitofrontal cortex, and ventral striatum (Dobryakova et al., 2017 [↗](#); Hernandez Lallement et al., 2014 [↗](#); Wagner et al., 2013 [↗](#)). Further evidence comes from studies focusing on the reward positivity (RewP) of the event-related potential (ERP) component. The RewP, a reliable neural signature of reward sensitivity, has its neural sources in the anterior cingulate cortex (Hauser et al., 2014 [↗](#)). Some studies found that RewP amplitude increased following the exertion of high-effort versus low-effort behavior for reward feedback (Bogdanov et al., 2022 [↗](#); Harmon-Jones et al., 2024 [↗](#); Ma et al., 2014 [↗](#); Umemoto et al., 2023 [↗](#); Yi et al., 2020 [↗](#)). While this framework has been used extensively to understand how effort expenditure influences reward processing during self-benefiting behaviors, surprisingly, it has not yet been applied to understand the reward after-effect of effort expenditure during prosocial acts.

In this study, we aimed to investigate the reward after-effect of prosocial effort, focusing on the RewP elicited during reward evaluation after effort expenditure. Previous studies examining how people vicariously process others’ rewards suggest a critical role of the RewP in prosocial behavior (Kwak et al., 2020 [↗](#); San Martín et al., 2016 [↗](#)). In these experiments, participants performed simple gambling or speeding reaction-time tasks to win monetary rewards for themselves and others. A consistent finding is that the RewP is attenuated for reward feedback benefiting others relative to oneself, including charity programs (San Martín et al., 2016 [↗](#)), beneficiaries with closer social distance (Kwak et al., 2020 [↗](#)), or anonymous individuals (Li et al., 2022 [↗](#)). However, rewards in these studies were earned with minimal effort. Crucially, previous research has never considered the impact of prior effort costs.

To address this issue, we recorded the RewP in a prosocial effort task where participants exerted varying levels of physical effort to earn monetary rewards for themselves or an anonymous other person. We hypothesized an effort-discounting effect on the RewP when exerting effort for others,

indicated by a reduced RewP as required effort increased. Conversely, we expected an effort-enhancement effect on the RewP when participants put in effort for themselves, shown by a more positive RewP as required effort increased. Further, we predicted that this dissociation would be contingent upon reward system activation. To provide a comprehensive characterization of prosocial effort, we subsequently examined participants' decision-making tendencies in a prosocial decision-making task where they chose to exert effort to benefit either themselves or others. We hypothesized that participants would exhibit reduced willingness to invest effort for others (i.e., prosocial apathy), and this behavioral tendency would modulate the neural after-effects of effort exertion.

2 Results

In this study, participants completed a role assignment task (Figure 1A) to become the decider in two subsequent prosocial tasks: a prosocial effort task and a prosocial decision-making task. In the prosocial effort task (Figure 1B), participants exerted five levels of physical effort (2–6 levels; Figure 1C) to earn rewards of varying amounts (¥0.2, ¥0.4, ¥0.6, ¥0.8, or ¥1.0) for themselves or an anonymous other person. The five effort levels were fully crossed with the five reward magnitudes, creating 25 unique combinations. Following the effort task, participants completed a prosocial decision-making task (Figure 1D) where they chose between a high-effort option (more effort for a larger reward) and a baseline option (no effort for a smaller reward). The high-effort options comprised the same 25 unique effort-reward combinations presented in the prosocial effort task. In both tasks, half of the trials benefited participants themselves, while the other half benefited others. We measured participants' neural responses to rewards obtained following effort exertion for themselves and others in the prosocial effort task, as well as their willingness to exert prosocial effort in the decision-making task. Additionally, to validate the manipulation, we collected participants' ratings of perceived difficulty, invested effort, and liking for each effort level across both beneficiary conditions.

2.1 Investing effort for others is less motivating than for self in the prosocial effort task

The mean maximum effort level (i.e., the average button-press count across three 6000-ms trials; see details in *Procedure*) was 36.06 ($SD = 4.95$, range = 26.33–46.00; Figure 2A). Participants achieved comparably high success rates in both self-benefiting ($M = 97\%$) and other-benefiting trials ($M = 96\%$). A mixed-effects logistic regression model of response success indicated that success rates decreased as effort demands increased ($b = -4.77$, $p < 0.001$). However, no other effects reached significance ($ps > 0.245$). Regarding response speed (button presses per second), participants responded faster as required effort increased ($b = 0.56$, $p < 0.001$; Figure 2B), but their speed did not vary with prospective reward magnitude ($b = 0.01$, $p = 0.128$; Figure 2C). Notably, participants also responded faster for themselves than for others ($b = -0.07$, $p < 0.001$). Full regression coefficients for the response success and speed models are provided in Supplementary Table S1.

To validate our experimental manipulation, we examined participants' subjective ratings as a function of effort level and beneficiary using linear mixed-effect regression models. As expected (Figure 2D), participants rated higher effort levels as more difficult ($b = 2.38$, $p < 0.001$), more demanding ($b = 1.90$, $p < 0.001$), and less likable ($b = -1.79$, $p < 0.001$). They felt less effort ($b = -0.32$, $p = 0.019$) and more disliking ($b = -0.62$, $p = 0.001$) for other-benefiting trials compared to self-benefiting trials. However, they perceived no differences in task difficulty between self-benefiting and other-benefiting trials ($b = 0.19$, $p = 0.260$). Moreover, the liking rating data exhibited a significant interaction between recipient and effort ($b = -0.28$, $p = 0.023$). Follow-up simple slopes analyses revealed that the discounting effect of effort on the liking rating was more pronounced when the beneficiary was others ($b = -1.93$, 95% CI = [-2.28, -1.58], $p < 0.001$) compared to when it was themselves ($b = -1.65$, 95% CI = [-2.00, -1.30], $p < 0.001$). The full regression estimates for rating data are provided in Supplementary Table S2. Together, our rating and behavioral data suggest

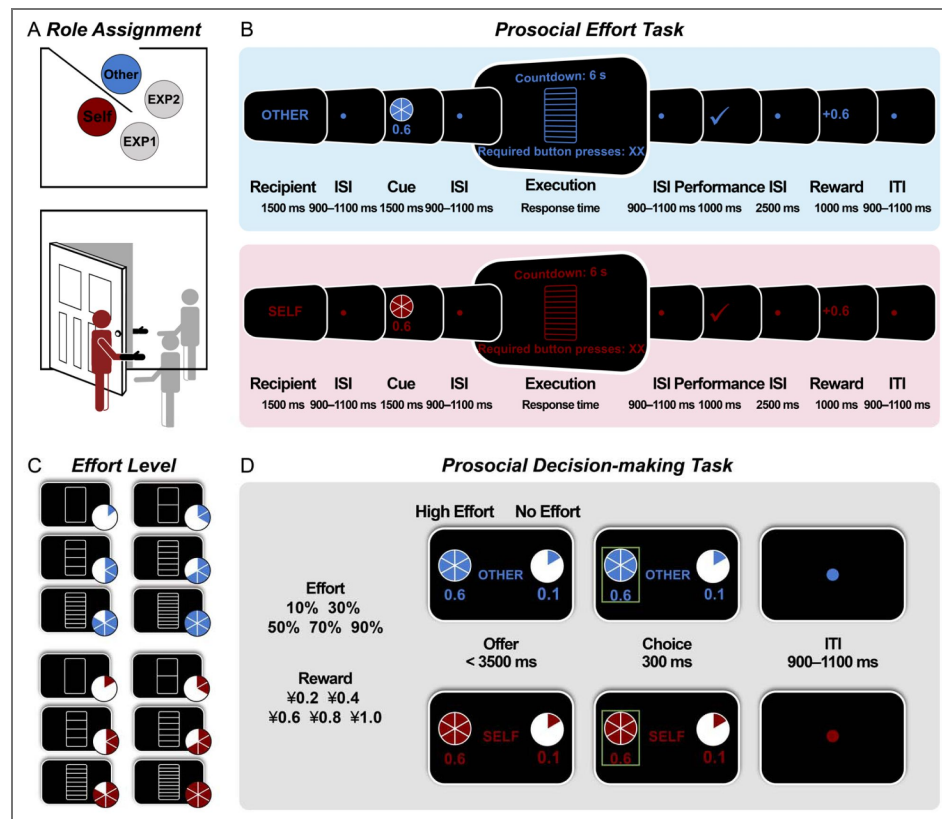


Figure 1. Experimental tasks.

(A) The role assignment task. Participants were introduced to another anonymous person and designated as a decider to invest physical effort for monetary rewards for themselves and others. (B) The prosocial effort task. Participants exerted physical effort (2–6 levels) to earn a potential reward of varying amounts (¥0.2, ¥0.4, ¥0.6, ¥0.8, or ¥1.0) for themselves and others. Successful effort had a 50% chance of yielding a reward. (C) Effort levels. The physical task required participants to rapidly press a button with their non-dominant pinky finger within 6000 ms. Effort level was visualized as the height of a vertical bar (10%, 30%, 50%, 70%, or 90% of the participant's calibrated maximum effort). The blank bar indicated no effort. (D) The prosocial decision-making task. Participants chose between a high-effort option (more effort for a larger reward) and a baseline option (no effort for a smaller reward). ISI = interstimulus interval; ITI = intertrial interval.

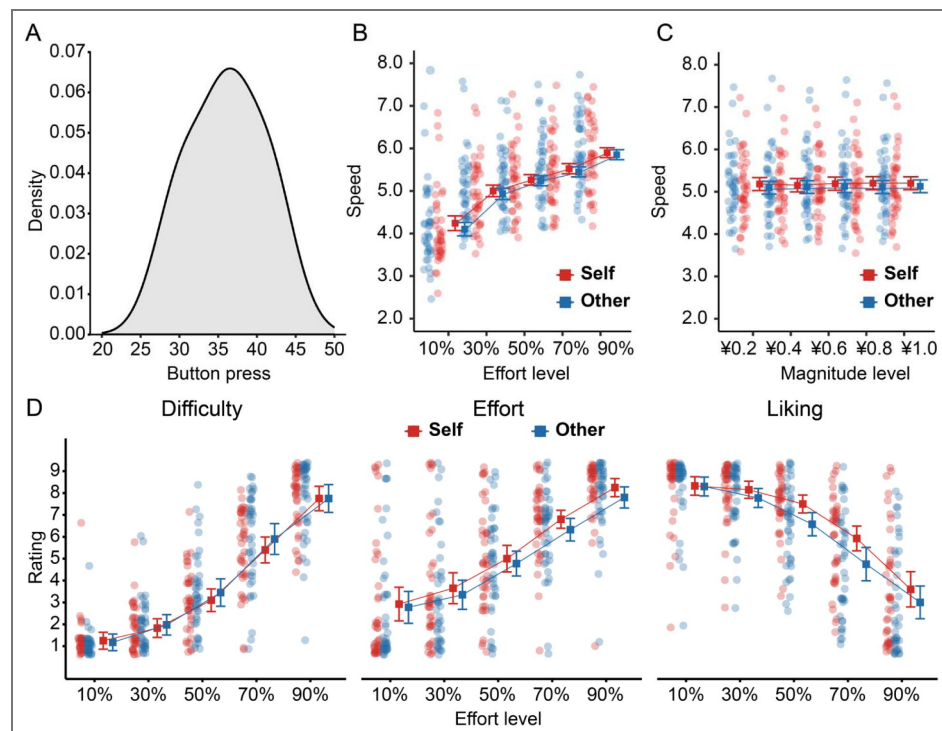


Figure 2. Behavioral and rating results of the prosocial effort task.

(A) Distribution of the maximum effort level (i.e., the average button-press count across three 6000-ms trials) across participants. (B–C) Response speed (button presses per second) data. Participants responded faster for themselves than for others. Response speed also increased as effort demands increased. (D) Rating data. Participants felt less effort and more disliking when exerting effort for others than for themselves. Error bars represent the within-subject standard error of the mean.

that the effort manipulation was successful. Furthermore, participants were less motivated to invest effort for others than for themselves at both subjective and behavioral levels, despite similar success rates.

2.2 Effort adds reward value for self but discounts reward value for others in the prosocial effort task

In the prosocial effort task, the RewP was evident as a relative positivity over frontocentral areas (Figure 3). We fitted the amplitude of the RewP using a linear mixed-effects regression model with recipient, effort, magnitude, valence, and their interactions as predictors. The full regression estimates for RewP data are shown in Figure 4A. As predicted, the RewP was significantly larger for self-benefiting relative to other-benefiting trials ($b = -0.68$, $p = 0.012$) and for gain versus nongain feedback ($b = -1.08$, $p < 0.001$), and became more positive as reward magnitude increased ($b = 0.42$, $p = 0.003$).

We observed a significant three-way interaction among recipient, magnitude, and valence ($b = 0.86$, $p = 0.038$). Subsequent simple slopes analyses (Figure 4B) revealed distinct neural patterns based on the beneficiary. When the beneficiary was self, the RewP became more positive as reward magnitude increased for both gain feedback ($b = 0.75$, 95% CI = [0.31, 1.19], $p < 0.001$) and nongain feedback ($b = 0.46$, 95% CI = [0.02, 0.90], $p = 0.039$). In contrast, when the beneficiary was others, the RewP became more positive as reward magnitude increased for nongain feedback ($b = 0.51$, 95% CI = [0.07, 0.95], $p = 0.022$), but not for gain feedback ($b = -0.02$, 95% CI = [-0.46, 0.42], $p = 0.923$).

Crucially, we observed a significant two-way interaction between recipient and effort ($b = -0.55$, $p = 0.009$), which was further qualified by a significant three-way interaction among recipient, effort, and magnitude ($b = -0.49$, $p = 0.019$). The predicted effects of the three-way interaction among recipient, effort, and magnitude are shown in Figure 4C. Visual inspection suggested that when reward magnitude was low, the RewP was insensitive to invested effort across self-benefiting and other-benefiting trials. However, as reward magnitude increased, the effect of effort expenditure on the RewP diverged depending on the beneficiary. To confirm these observations statistically, we conducted post-hoc simple slopes analyses at 1 standard deviation (SD) below (“Low”) and above (“High”) the mean reward magnitude. At low reward magnitude, the RewP did not vary with effort level, regardless of whether the beneficiary was self ($b = 0.01$, 95% CI = [-0.42, 0.44], $p = 0.958$) or others ($b = -0.05$, 95% CI = [-0.48, 0.39], $p = 0.829$). In contrast, a striking dissociation emerged at high reward magnitude. When the beneficiary was self, the RewP tended to be more positive as effort level increased ($b = 0.42$, 95% CI = [-0.01, 0.85], $p = 0.056$), suggesting an effort-enhancement effect. Conversely, when the beneficiary was others, the RewP became less positive as effort level increased ($b = -0.63$, 95% CI = [-1.05, -0.20], $p = 0.004$), indicating an effort-discounting effect.

To establish the specificity of our RewP findings, we examined the parietal P3 in response to performance feedback (i.e., effort-completion cues; see Supplementary Figure S1 for the ERP waveforms and topographic maps) using a linear mixed-effect regression model with recipient, effort, magnitude, and their interactions as predictors. We found a main effect of effort on the P3 in response to effort-completion cues, with its amplitudes increasing with prior effort levels ($b = 0.72$, $p < 0.001$). Importantly, this sensitivity to effort was invariant across beneficiary conditions, as indicated by a nonsignificant interaction between recipient and effort ($b = -0.19$, $p = 0.285$). No other significant effects were observed (see Supplementary Table S3 for full regression estimates).

2.3 Reward is devalued by effort to a higher degree for others than for self in the prosocial decision-making task

Decision times in the prosocial decision-making task (Figure 5A–B) were fitted using a linear mixed-effects regression model with recipient, magnitude, and both linear and quadratic terms for effort, along with their interactions, as predictors. The analysis yielded significant main effects of effort (linear: $b = 38.91$, $p = 0.003$; quadratic: $b = -64.89$, $p < 0.001$) and magnitude ($b = -54.88$, $p <$

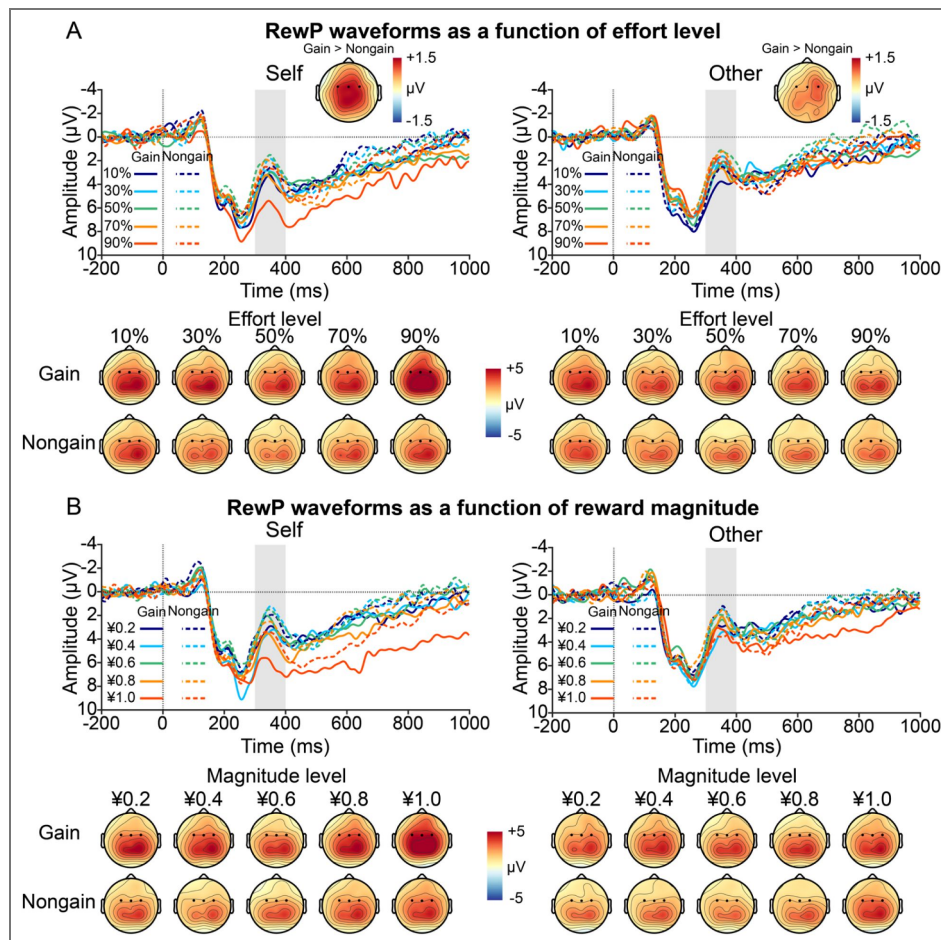


Figure 3. Grand-average ERP waveforms and topographic maps of the RewP as a function of recipient (self vs. other) and valence (gain vs. non-gain) separately for effort (A) and reward (B) trials.

Gray shaded bars represent time windows used for quantification.

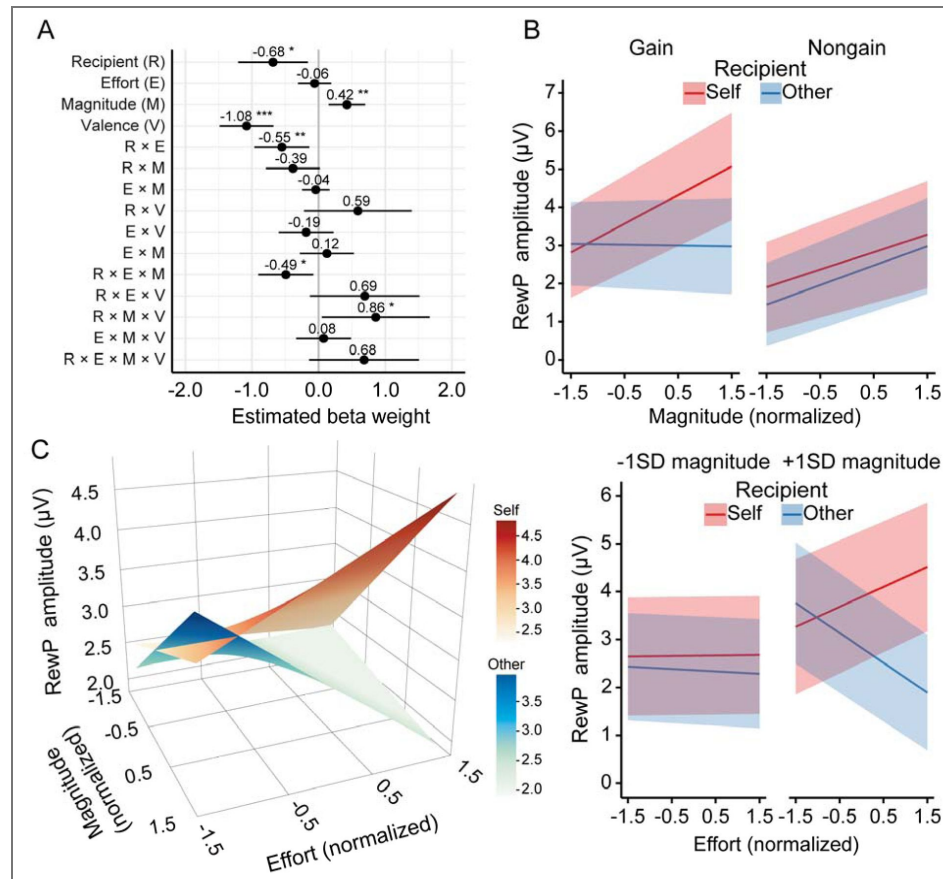


Figure 4. RewP results in the prosocial effort task.

(A) Estimated beta weights for the mixed-effects model predicting RewP amplitudes. The RewP model was specified as: Amplitude ~ Recipient * Effort * Magnitude * Valence + (Recipient + Effort + Magnitude | Participant). (B) Fixed effects of reward magnitude on the RewP as a function of recipient and valence during reward evaluation, showing a significant three-way interaction. (C) Fixed effects of effort and reward on the RewP as a function of recipient during reward evaluation. The left graph displays the fixed effects with effort and reward as continuous predictors, whereas the right graph shows fixed effects of effort at 1 SD below and above the mean reward magnitude. An effort-enhancement effect emerged when participants invested effort for themselves, whereas an effort-discounting effect occurred when they exerted effort for others. This dissociable after-effect was present only when reward magnitude was high. Shaded areas depict 95% confidence intervals. * $p < .05$, ** $p < .01$, *** $p < .001$.

0.001). Crucially, we observed significant interactions between recipient and effort ($b = -41.49$, $p = 0.001$), as well as between recipient and magnitude ($b = 44.86$, $p = 0.020$). Simple slopes analyses indicated that decision times increased with effort in self-benefiting trials ($b = 59.70$, 95% CI = [32.87, 86.50], $p < 0.001$) but not in other-benefiting trials ($b = 18.30$, 95% CI = [-8.55, 45.10], $p = 0.182$). Similarly, higher reward magnitude facilitated response speed more substantially in self-benefiting trials ($b = -77.30$, 95% CI = [-105.70, -48.90], $p < 0.001$) than in other-benefiting trials ($b = -32.50$, 95% CI = [-60.80, -4.20], $p = 0.024$). Furthermore, both the linear and quadratic effects of effort were qualified by significant interactions with reward magnitude (linear interaction: $b = 33.23$, $p < 0.001$; quadratic interaction: $b = 15.14$, $p = 0.040$). Simple slopes analyses of the linear term showed that decision times increased with effort at high reward magnitude ($M + 1SD$: $b = 72.10$, 95% CI = [45.30, 98.90], $p < 0.001$) but not at low reward magnitude ($M - 1SD$: $b = 5.61$, 95% CI = [-21.20, 32.40], $p = 0.682$). Decomposition of the quadratic interaction revealed that the non-linear curvature of the effort effect was more pronounced at low reward magnitude ($M - 1SD$: $b = -160.10$, 95% CI = [-218.30, -101.90], $p < 0.001$) than at high reward magnitude ($M + 1SD$: $b = -99.50$, 95% CI = [-157.60, -41.40], $p = 0.001$). Full regression estimates for decision-time data are provided in [Supplementary Table S4](#).

We fitted participants' choice behavior using a mixed-effects logistic regression model with recipient, effort, magnitude, and their interactions as predictors. As depicted in [Figure 5C–D](#), participants were less willing to invest effort as effort levels increased ($b = -2.54$, $p < 0.001$) and more willing as reward magnitude increased ($b = 2.11$, $p < 0.001$). They were also less willing to exert effort for others than for themselves ($b = -1.74$, $p < 0.001$). This recipient effect was modified by a significant interaction between recipient and magnitude ($b = -0.56$, $p < 0.001$). Simple slopes analyses revealed that the incentive effect of reward on willingness to exert effort was less pronounced when the beneficiary was another person ($b = 1.83$, 95% CI = [1.36, 2.31], $p < 0.001$) compared to when it was participants themselves ($b = 2.39$, 95% CI = [1.89, 2.90], $p < 0.001$). We also found a significant interaction between effort and magnitude ($b = 0.15$, $p = 0.028$). Follow-up simple slopes analyses revealed that the discounting effect of effort on willingness was more pronounced at low reward magnitude ($M - 1SD$: $b = -2.69$, 95% CI = [-3.09, -2.29], $p < 0.001$) than at high reward magnitude ($M + 1SD$: $b = -2.38$, 95% CI = [-2.82, -1.94], $p < 0.001$). Full regression estimates for choice data are shown in [Supplementary Table S5](#). These findings were further corroborated by a parabolic discounting model (see Materials and methods). As shown in [Figure 5E](#), participants exhibited a higher (i.e., steeper) discounting rate for other-benefiting choices ($M = -3.78$) compared to self-benefiting choices ($M = -4.48$, $p < 0.001$). Furthermore, a higher discounting rate for others was associated with a higher discounting rate for oneself ($r = 0.52$, $p < 0.001$), suggesting a domain-general motivation that influences effort discounting across beneficiary contexts.

2.4 Neural after-effects of prosocial effort from the prosocial effort task are modulated by effort discounting rate from the effort decision-making task

To bridge our neural findings with behavioral preferences, we examined whether individual differences in effort discounting (quantified by the discounting parameter K) from the prosocial decision-making task modulated the neural after-effects of effort (as indexed by the RewP) in the prosocial effort task. To this end, we added participants' K values (log-transformed and z-scored) to the above-described RewP model as a fixed effect predictor, allowing it to interact with all other predictors of interest. Full regression estimates for the resultant model are provided in [Supplementary Table S6](#).

We observed a significant four-way interaction among recipient, effort, magnitude, and K value ($b = 0.58$, $p = 0.013$). Crucially, this higher-order interaction was robust, as it was replicated when we replaced individual K values with high-effort choice proportions from the prosocial decision-making task ($b = -0.58$, $p = 0.014$; see [Supplementary Tables S6](#)). To decompose this complex interaction, we performed simple slopes analyses separately for self- and other-benefiting trials at high and low levels of reward magnitude and discounting rate ($\pm 1SD$). As shown in [Figure 6](#), for

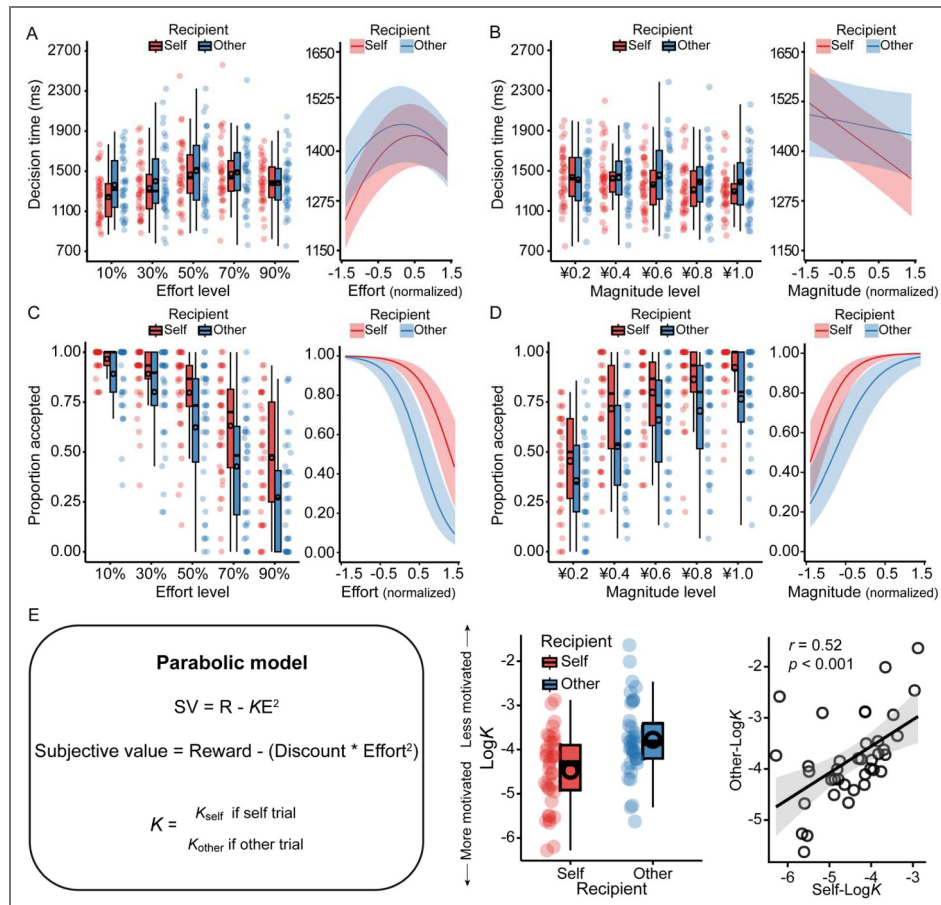


Figure 5. Behavioral and discounting results of the prosocial decision-making task.

(A–B) Participants took longer to make decisions as effort level increased in self trials but not in other trials (A). Increased reward magnitude decreased the decision time more pronouncedly in self trials than in other trials (B). (C–D) Participants were less willing to invest effort for others than for themselves. (E) Effort exertion discounted rewards to a higher degree when the beneficiary was others compared to when it was themselves (left and middle). A higher discounting rate for others was associated with a higher discounting rate for self (right). The black circles overlaid on the boxplots indicate the mean across participants. Shaded areas depict 95% confidence intervals. Noted that seven subjects had an accuracy rate of less than 60% on catch trials, but this did not influence the results of the prosocial decision-making task. Moreover, two participants had one negative original K value.

self-benefiting trials, the effort-enhancement effect on the RewP was significant only for participants with high discounting rates at low reward magnitude ($b = 1.02$, 95% CI = [0.22, 1.82], $p = 0.012$). In contrast, participants with low discounting rates exhibited no significant effort effect ($b = -0.37$, 95% CI = [-0.89, 0.15], $p = 0.159$). At high reward magnitude, simple slopes analyses detected no significant effort effects for either high ($b = 0.35$, 95% CI = [-0.44, 1.14], $p = 0.383$) or low ($b = 0.45$, 95% CI = [-0.07, 0.97], $p = 0.093$) discounting individuals. For other-benefiting trials, participants with low discounting rates exhibited a significant effort-discounting effect at high reward magnitude ($b = -0.97$, 95% CI = [-1.74, -0.20], $p = 0.014$). In contrast, no significant effort effects were observed for participants with high discounting rates at either high ($b = -0.45$, 95% CI = [-0.97, 0.08], $p = 0.098$) or low ($b = -0.16$, 95% CI = [-0.69, 0.38], $p = 0.564$) reward magnitudes, nor for participants with low discounting rates at low reward magnitude ($b = 0.14$, 95% CI = [-0.64, 0.92], $p = 0.729$).

3 Discussion

Many prosocial behaviors require effort investment, yet people are often reluctant to exert effort for others' benefit relative to their own. In this study, we examined how effort expenditure influences subsequent reward evaluation during prosocial acts, addressing a critical gap in previous research. We found dissociable reward after-effects of effort exertion between self-benefiting and other-benefiting acts: prior effort potentiated reward evaluation when the beneficiary was oneself but attenuated it when the beneficiary was another person. Moreover, this dissociation occurred specifically when the potential reward magnitude was large and was modulated by individual differences in effort discounting, yet remained independent of performance evaluation.

In our study, despite perceiving no differences in task difficulty and achieving comparable success rates for themselves and others, participants took longer to exert effort, reported less effort, and disliked the task more as the required effort increased when helping others. Participants exhibited a higher discounting rate for choices benefiting others versus those benefiting themselves, and their decision time was less influenced by effort and reward levels when the beneficiary was others compared to when it was themselves. These findings are largely consistent with previous studies, demonstrating a prosocial apathy when physical effort is required to help others (Contreras-Huerta et al., 2020 [DOI](#)).

Our most important finding is the dissociation of the reward after-effect of effort expenditure between self-benefiting and other-benefiting trials. When the beneficiary was oneself, the RewP was potentiated as prior invested effort increased. In contrast, when the beneficiary was others, the RewP was attenuated as prior invested effort increased. Given that the RewP is considered a reliable neural signature of reward sensitivity (Proudfit, 2015 [DOI](#)), our data suggest an effort-enhancement effect for self-benefiting acts but an effort-discounting effect for other-benefiting acts. The effort-enhancement effect aligns with previous studies where effort is exerted to obtain rewards for oneself (Bogdanov et al., 2022 [DOI](#); Ma et al., 2014 [DOI](#); Umemoto et al., 2023 [DOI](#); Yi et al., 2020 [DOI](#)). Several theories have been proposed to interpret this effect. For instance, prior effort expenditure may enhance the subjective value of a reward through cognitive dissonance reduction (i.e., effort justification) to alleviate the psychological discomfort resulting from having engaged in unpleasant effort (Aronson & Mills, 1959 [DOI](#)) or via the psychological contrast between the aversive state elicited by effort expenditure and the reward that follows (Zentall, 2010 [DOI](#)). While these perspectives adequately explain the effort-enhancement effect for self-benefiting acts, they fall short in explaining the discounting effect observed here for other-benefiting acts. According to these views, the more aversive state elicited by investing effort for others should lead to stronger justification or a more intense psychological contrast, which consequently should increase the reward value to a greater degree. However, we observed an effort-discounting effect when it came to prosocial acts. This discounting effect may result from the heightened salience of opportunity costs in helping others, that is, the value of the next-best use of the effort devoted to the current task (Kurzban et al., 2013 [DOI](#)).

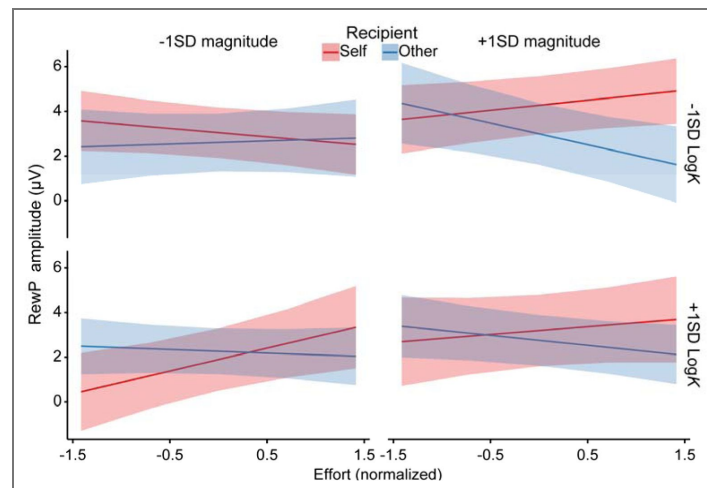


Figure 6. Cross-task modulation results.

Individual differences in effort discounting (log-transformed K) estimated from the prosocial decision-making task modulated the neural after-effects of effort exertion in the prosocial effort task. For self-benefiting trials, high-discounting individuals exhibited an effort-enhancement effect on the RewP specifically at low reward magnitude; conversely, for other-benefiting trials, low-discounting individuals exhibited an effort-discounting effect only at high reward magnitude. Fixed effects are visualized at ± 1 SD from the mean.

Interestingly, the dissociable after-effect of effort expenditure occurred only when reward magnitude was high. When reward magnitude was low, prior invested effort did not affect the RewP across self-benefiting and other-benefiting trials. These results suggest that the dissociable effort after-effect relies on the involvement of the motivational system, which might be activated only when potential reward is high. This possibility was further supported by our P3 data in response to effort-completion cues during performance evaluation. Specifically, participants exhibited an increased P3 when seeing a feedback stimulus informing them that the required effort has been achieved, whether the beneficiary was themselves or others. As the P3 is thought to reflect motivational salience based on feedback evaluation (Nieuwenhuis et al., 2005 [↗](#)), our results suggest that participants could derive value from their successful effort completion independently of the reward outcome. Unlike reward evaluation, the effort-enhancement effect on the P3 is associated with intrinsic motivation such as pride and achievement stemming from successful performance (Bowyer et al., 2021 [↗](#); Jiang & Zheng, 2023 [↗](#)). Together, our data establish that the dissociation in the after-effect of prosocial effort occurs only when reward motivation is sufficiently activated.

While previous research on prosocial effort has focused exclusively on cognitive processes before and during effort expenditure (Contreras-Huerta et al., 2020 [↗](#)), our findings provide a comprehensive picture of prosocial effort. We not only replicated the greater effort discounting for prosocial acts compared to self-benefiting acts but also demonstrated a dissociable after-effect of prosocial effort. Our results suggest that the difference between self-benefiting and other-benefiting effort is quantitative before and during effort expenditure but qualitative after effort expenditure. This after-effect of prosocial effort, together with the established discounting effect before effort exertion, offers a new perspective for facilitating prosocial behavior. When working for oneself, effort expenditure not only prospectively discounts but also retrospectively increases the subjective value of reward, a phenomenon referred to as the *effort paradox* (Inzlicht et al., 2018 [↗](#)). Crucially, our cross-task analysis reveals that this paradox is not universal but is driven specifically by individuals highly sensitive to effort costs (high K), particularly when external rewards are low. This finding strongly supports a cognitive dissonance account: those who find effort most aversive are most compelled to inflate the value of small rewards to justify their exertion (Aronson & Mills, 1959 [↗](#)).

In contrast, for prosocial acts, this justification mechanism appears absent. Instead, we observed a persistent effort discounting before, during, and after effort expenditure, which was most pronounced in individuals with low effort sensitivity (low K) when reward magnitude was high. This seemingly paradoxical pattern might be interpreted through the lens of disadvantageous inequity aversion (Fehr & Schmidt, 1999 [↗](#)). Specifically, the combination of high personal effort and high monetary reward for another person creates a salient disparity between the participant's incurred cost and the recipient's gain. Although low- K individuals are behaviorally willing to tolerate this cost, their neural valuation system may nonetheless track the “unfairness” of this asymmetry, thereby attenuating the neural reward signal (Tricomi et al., 2010 [↗](#)). These insights suggest that facilitating prosocial behavior may require not just lowering costs, but potentially framing outcomes to trigger the effort justification mechanisms that drive the effort paradox observed in self-benefiting acts (Inzlicht & Campbell, 2022 [↗](#)). Consistent with this view, a promising direction for future studies is to manipulate the social distance of the recipient. This approach will help determine whether the after-effect of effort expenditure could shift from an effort-discounting effect to an effort-enhancement effect when the beneficiary is a close other, such as a friend or family member, thereby bridging the gap between self- and other-oriented motivation (Jones & Rachlin, 2006 [↗](#)).

One limitation of this study concerns the potential confound between temporal delay and effort level. Because higher effort levels required more time to complete, effort expenditure might not have directly affected subsequent reward evaluation but instead increased participants' delay discounting. If this were the case, one would expect a similar discounting effect on the RewP for self-benefiting trials, as the RewP amplitude typically decreases as the time to receive a reward increases (Zheng et al., 2023 [↗](#)). Moreover, temporal delay is unlikely to account for our findings

given the relatively small difference in duration between the lowest and highest effort levels in our task (Weinberg et al., 2012 [↗](#)). Nonetheless, future work is needed to strictly control for duration to rule out this possibility

In conclusion, our results demonstrate that the neural evaluation of reward outcomes is dynamically shaped by the prior cost of effort in a beneficiary-dependent manner. We identified a critical qualitative dissociation: whereas self-benefiting effort paradoxically enhances reward valuation, other-benefiting effort induces a persistent reward devaluation. Crucially, this divergence represents a distinct valuation process that emerges specifically at high reward magnitude, is modulated by individual differences in effort discounting, and operates independently of performance evaluation. These findings imply that the *effort paradox* where labor adds value is restricted to self-benefiting acts. Consequently, promoting prosocial behavior may require not just reducing physical costs, but reframing prosocial outcomes to trigger the justification mechanisms that naturally support self-benefiting exertion.

4 Materials and methods

All data and code used for this study are available on OSF at <https://osf.io/bvpa2/> [↗](#). This study was not preregistered.

4.1 Participants

Forty-seven right-handed university students were recruited for this study through local advertisements. Seven participants were excluded from data analysis due to the following reasons: five never chose to work on either self-benefiting or other-benefiting trials, one had a success rate of 35% on the maximum effort level during the prosocial effort task, and one doubted the impact of their actions on the other participant. The final sample consisted of 40 participants (20 females; $M = 21.55$ years, $SD = 2.65$). We performed a sensitivity analysis using the *simr* v1.0.6 package (Green & MacLeod, 2016 [↗](#)) to compare the regression weight for each effect of interest with the smallest detectable effect size at a power of 80% based on the current sample. The results showed that most of significant effects observed were larger than the smallest detectable effect, suggesting that our sample size provided adequate statistical power. All participants had normal or corrected-to-normal vision and reported no psychiatric or neurological disorders. They received ¥25 for participation and a bonus of ¥29–¥37 based on their task performance. Each participant provided written informed consent, and this study was approved by a local Institutional Review Board.

4.2 Procedure

Upon arrival at the lab, participants undertook a role assignment task, leading them to believe that they would complete two EEG tasks with a partner. Afterwards, they performed a prosocial effort task and a prosocial decision-making task while their EEG was recorded. Following the EEG tasks, participants used a 9-point Likert scale (ranging from 1 = not at all to 9 = very much) to rate their perceived difficulty, invested effort, and liking at each effort level when exerting effort for themselves and others during the tasks.

The role assignment task

This task was adapted from a previous study (Lockwood et al., 2017 [↗](#)). Participants were led to one side of a door and informed that a second participant (who was in fact a confederate) was also involved in this study (Figure 1A [↗](#)). The confederate, of the same gender as the participant, was then escorted by a different experimenter to the opposite side of the door. Both the participant

d confederate were handed a black glove and instructed not to speak for anonymity. They acknowledged each other's presence by waving their gloved hands, without ever being seen by each other. The experimenter then tossed a coin to determine who would choose a ball from a box first. Their roles (a receiver vs. a decider) were assigned based on the outcome. Unbeknownst to participants, they were always designated as the decider, responsible for performing tasks for themselves and others. The confederate was assigned the role of the receiver, responsible solely for completing tasks for themselves. To prevent potential effects of social norms such as reciprocity (Gintis et al., 2003 [DOI](#)), both the participant and confederate were informed that they would not be aware of each other's performance and would leave the lab at different times.

The prosocial effort task

This task was designed to measure participants' neural responses to rewards obtained after investing physical effort for themselves and others (Figure 1B [DOI](#)). Before the task, participants completed three trials in which they were asked to press a button as rapidly as possible with their non-dominant pinky finger for 6000 ms. The maximum effort level was operationalized as the average button-press count across the three trials. In the prosocial effort task, each trial began with a cue for 1500 ms, indicating whether it was self-benefiting or other-benefiting. Participants exerted physical effort to earn rewards for themselves in self-benefiting trials and for others in other-benefiting trials. Self and other trials were highlighted in different colors (blue and red) throughout the trial, which was counterbalanced across participants. After a jittered interval (900–1100 ms), a pie chart with a number below it appeared for 1500 ms, showing the required effort level (10%, 30%, 50%, 70%, or 90% of their maximum effort level) and the potential reward (¥0.2, ¥0.4, ¥0.6, ¥0.8, or ¥1.0), respectively (Figure 1C [DOI](#)). The five effort levels were fully crossed with the five reward magnitudes, creating 25 unique combinations. Following another jittered interval (900–1100 ms), participants entered the effort-execution phase, making the required button presses with their non-dominant pinky finger within 6000 ms. After this phase and another jittered interval (900–1100 ms), a feedback stimulus was shown for 1000 ms, indicating whether the required effort level was achieved. If participants succeeded, a tick was presented, signaling that they were eligible to win the cued reward. If they failed, a cross was displayed, followed by a new trial after a jittered interval of 900–1100 ms. After a 2500-ms interval, a feedback stimulus was displayed for 1000 ms, indicating whether participants received the reward or not. Gains and nongains were equally likely and delivered pseudorandomly. Each trial ended with an interval varying between 900 and 1100 ms. The task consisted of 200 trials (100 for self-benefiting and 100 for other-benefiting, respectively) divided into eight blocks of 25 trials, with a self-determined break between blocks. Eight practice trials familiarized participants with the task before the formal experiment.

The prosocial decision-making task

This task was designed to measure participants' willingness to exert effort for themselves and others (Figure 1D [DOI](#)). Participants made decisions between a baseline no-effort option for ¥0.1 and a high-effort option for a greater reward. The high-effort options were the same as those in the prosocial effort task, including 25 unique effort-reward combinations. Participants had 3500 ms to decide using their left or right index finger. The chosen option was highlighted with a green border for 300 ms. Failure to respond within the time limit resulted in ¥0 and a 1000-ms warning message of "Please respond within 3500 ms". Each trial ended with a jittered interval of 900–1100 ms. The task consisted of 150 trials, with each high-effort option occurring six times. Half of the trials benefited participants themselves, while the other half benefited others. To ensure that participants stayed focused on the task, we included 20 catch trials (10 for each beneficiary condition) where participants confirmed the effort and reward levels of the previous high-effort option. To ensure that our task was incentive-compatible, participants were told that they would complete their chosen effort option on 16 randomly selected trials (8 for each beneficiary condition) to determine their final reimbursement. They were instructed to consider each decision carefully because each trial choice would be selected. Before the experiment, participants

completed 10 practice trials for familiarization. After the task, participants exerted their effort required for their choices in the 16 selected trials. Finally, they were asked to report whether they believed they were earning rewards for the other participant (i.e., the receiver).

4.3 EEG recording and processing

EEG data were recorded using 28 Ag/AgCl channels placed on an elastic cap based on the international 10–20 system. Two additional channels were positioned on the left and right mastoids. The EEG were recorded with a reference channel placed between Cz and CPz. Horizontal and vertical electrooculograms were recorded from two pairs of channels over the external canthi of each eye and the left suborbital and supraorbital ridges, respectively. EEG signals were amplified using a Neuroscan Graef 4K amplifier with a low-pass filter of 100 Hz in DC acquisition mode and digitized at a rate of 512 samples per second. Channel impedances were maintained below 5 K Ω .

The EEG data were analyzed using EEGLAB v2021.0 (Delorme & Makeig, 2004 [DOI](#)) and ERPLAB v8.10 (Lopez-Calderon & Luck, 2014 [DOI](#)) toolboxes in MATLAB 2020b (MathWorks, US). The signals were rereferenced to the average of the left and right mastoids and filtered with a bandpass of 0.1–35 Hz using a zero phase-shift Butterworth filter (12 dB/octave roll-off). Channels with poor quality or excessive noise were interpolated using the spherical interpolation algorithm, and portions of EEG containing extreme voltage offsets or break periods were removed. Ocular artifacts were removed using an infomax independent component analysis on continuous EEG with the help of the ICLABEL algorithm (Pion-Tonachini et al., 2019 [DOI](#)). Epochs were then extracted from -200 to 1000 ms relative to feedback onset, with the prestimulus average activity as the baseline. An automatic artifact detection algorithm was applied to remove epochs with a voltage difference exceeding 50 μ V between sample points or 200 μ V within a trial, a maximum voltage difference less than 0.5 μ V within 100-ms intervals, or a slow voltage drift with a slope greater than $\pm$ 100 μ V. On average, 97.71% of trials were retained for statistical analysis. Single-trial RewP amplitude was measured as mean voltage from 300 to 400 ms relative to reward feedback onset (i.e., reward delivery) over frontocentral channels (FC3, FCz, FC4). We also measured the parietal P3 (300–440 ms; averaged across P3, Pz, and P4) in response to performance feedback (i.e., effort completion), given its relationship with motivational salience (Bowyer et al., 2021 [DOI](#); Ma et al., 2014 [DOI](#)). Measurement parameters (time windows and channel sites) were determined from the grand-averaged ERP waveforms and topographic maps collapsed across all conditions, which was thus orthogonal to the conditions of interest (Luck & Gaspelin, 2017 [DOI](#)). Data were averaged across the selected electrode clusters to improve signal-to-noise ratio and reliability.

4.4 Data analysis

Single-trial data were exported into R v4.2.2 for statistical analyses. Key statistical analyses utilized mixed-effects regression models with random intercepts and slopes (unstructured covariance matrix), implemented in the *lme4* package v1.1.31 (Bates et al., 2015 [DOI](#)). For the prosocial effort task, we analyzed response success data using a mixed-effects logistic regression model with a binomial link function. We analyzed response speed (i.e., button presses per second) and ERP data using linear mixed-effects regression models. These models included recipient, effort, magnitude, valence, and their interactions as predictors. For the prosocial decision-making task, we fitted decision time data using a linear mixed-effects regression model and choice data using a mixed-effects logistic regression model with a binomial link function.

The choice model included recipient, effort, magnitude, and their interactions as predictors. For the decision time model, visual inspection of the data (Figure 5A [DOI](#), left panel) suggested a non-linear relationship between effort and decision time. Therefore, we included both linear and quadratic terms for effort, along with recipient, magnitude, and their interactions, as predictors. We fitted post-experimental rating data separately using a linear mixed-effect regression model with recipient, effort, and their interactions as predictors. For all models, we contrast-coded categorical regressors (recipient: -0.5 for self and +0.5 for other; valence: -0.5 for gain and +0.5 for nongain) and z-scored continuous regressors (effort and magnitude) within participants. For each

model, we fitted the maximal random-effects structure and, when the model was overparameterized, used singular value decomposition to simplify the random-effects structure until the model converged. For linear mixed-effect models, *p*-values were calculated Satterthwaite's approximation for degrees of freedom. For mixed-effect logistic models, *p*-values were obtained using Wald Z-tests. To decompose significant interaction, we performed follow-up pairwise comparisons on estimated marginal means. We excluded trials with failed responses (3.34%) in the prosocial effort task and trials with no responses (0.91%) in the prosocial decision-making task from statistical analyses.

To quantify how participants devalued rewards by effort exertion for themselves and others, we fitted their choices in the prosocial decision-making task separately for self-benefiting and other-benefiting trials with a parabolic function:

$$SV = R - KE^2$$

Where *SV* represents the subjective net value of the high-effort option with a given reward (*R*) and effort (*E*). The discount parameter *K* characterizes the degree to which the reward is discounted by required effort. A higher *K* value indicates that the reward is devalued by the effort to a higher degree. The derived *SV* of the high-effort and baseline no-effort options were compared and transformed into the probability of choosing the high-effort option through a SoftMax function:

$$P_{high\ effort} = \frac{e^{\beta * SV_{high\ effort}}}{e^{\beta * SV_{no\ effort}} + e^{\beta * SV_{high\ effort}}}$$

where β represents the slope of the logistic function, which is a participant-specific parameter and reflects the sensitivity to *SV* differences between options (Collins & Shenhav, 2022 [↗](#)). The parabolic model with two separate *K* parameters for self and other trials has been validated in previous studies using a similar task (Lockwood et al., 2021 [↗](#); Lockwood et al., 2022 [↗](#)). We normalized the *K* values by taking the natural logarithm (i.e., $\log K$) and compared these $\log K$ values between self and other trials using a paired-*t* test.

Data availability

Data and code that support the findings of this study are available on Open Science Framework at <https://osf.io/bvpa2/> [↗](#).

Supplementary Materials

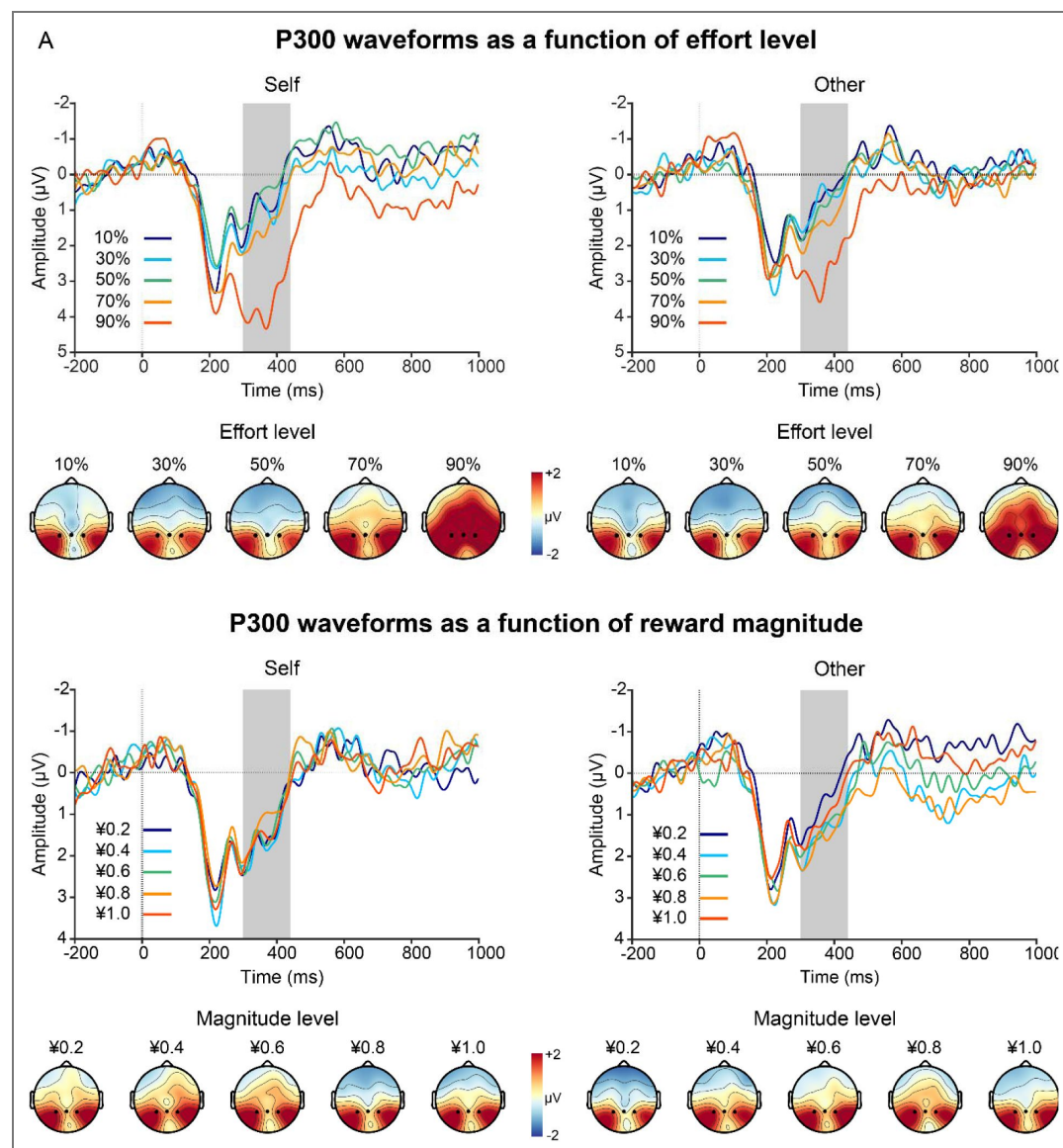


Figure S1. Grand-average ERP waveforms and topographic maps of the P3 as a function of recipient (self vs. other) separately for effort (A) and reward (B) trials. Gray shaded bars represent time windows used for quantification.

Table S1. Results of a linear mixed-effects model predicting response success (left) and response speed (right) in the prosocial effort task.

Predictors	Response success			Response speed		
	<i>b</i>	95% CI	<i>p</i>	<i>b</i>	95% CI	<i>p</i>
Intercept	8.71	6.79, 10.63	<0.001	5.15	4.91, 5.39	<0.001
Recipient (R)	-0.46	-2.34, 1.42	0.632	-0.07	-0.11, -0.04	<0.001
Effort (E)	-4.77	-6.14, -3.41	0.001	0.56	0.47, 0.64	<0.001
Magnitude (M)	0.45	-0.31, 1.20	0.246	0.01	-0.00, 0.02	0.128
R:E	0.07	-1.29, 1.42	0.923	0.02	-0.00, 0.05	0.074
R:M	0.40	-1.11, 1.90	0.606	-0.00	-0.03, 0.02	0.847
E:M	-0.24	-0.79, 0.31	0.400	0.01	-0.01, 0.02	0.276
R:E:M	-0.18	-1.28, 0.92	0.748	0.00	-0.03, 0.03	0.965
Observations	8000			8000		

Notes: The final model for response success data was specificized as Response success ~ Recipient * Effort * Magnitude + (Recipient + Effort | Participant) and for response speed data as: Response speed ~ Recipient * Effort * Magnitude + (Recipient + Effort | Participant). Both effort and magnitude levels were standardized before being entered into the model. For response success, *b* values are expressed in log-odds units from a logistic mixed-effects model, whereas for response speed, *b* values are estimates from a linear mixed-effects model. Statistically significant *p* values (< 0.05, two-sided) are shown in bold. CI = confidence interval.

Table S2. Results of linear regression models predicting rating data of difficulty, effort, and liking.

Predictors	Difficulty			Effort			Liking		
	<i>b</i>	95% CI	<i>p</i>	<i>b</i>	95% CI	<i>p</i>	<i>b</i>	95% CI	<i>p</i>
Intercept	3.96	3.66, 4.25	<0.001	5.17	4.63, 5.70	<0.001	6.39	6.00, 6.78	<0.001
Recipient (R)	0.19	-0.14, 0.51	0.260	-0.32	-0.58, -0.06	0.019	-0.62	-0.98, -0.26	0.001
Effort (E)	2.38	2.19, 2.57	<0.001	1.90	1.54, 2.26	<0.001	-1.79	-2.11, -1.46	<0.001
R:E	0.07	-0.16, 0.30	0.548	-0.11	-0.30, 0.09	0.270	-0.28	-0.51, -0.04	0.023
Observations	400			400			400		

Notes. The final model for rating data was specificized as: Rating ~ Recipient * Effort + (Recipient + Effort | Participant). Effort level was standardized before being entered into the model. Statistically significant *P* values (< 0.05, two-sided) are shown in bold. CI = confidence interval.

Table S3. Results of a linear mixed-effects model predicting P3 amplitudes in response to performance feedback in the prosocial effort task.

Predictor	<i>b</i>	95% CI	<i>p</i>
Intercept	1.31	0.74, 1.88	<0.001
Recipient (R)	-0.15	-0.48, 0.19	0.393
Effort (E)	0.72	0.45, 0.99	<0.001
Magnitude (M)	0.01	-0.16, 0.18	0.904
R:E	-0.19	-0.53, 0.16	0.285
R:M	0.17	-0.16, 0.51	0.319
E:M	-0.13	-0.30, 0.04	0.131
R:E:M	0.00	-0.34, 0.34	0.984
Observations	7469		

Notes. The final model was specificized as: Amplitude ~ Recipient * Effort * Magnitude + (Effort | Participant). Both effort and magnitude levels were standardized before being entered into the model. Statistically significant *p* values (< 0.05, two-sided) are shown in bold. CI = confidence interval.

Table S4. Results of a linear mixed-effects models predicting decision times in the prosocial decision-making task.

Predictors	<i>b</i>	95% CI	<i>p</i>
Intercept	1470.46	1373.63, 1567.29	<0.001
Recipient (R)	37.14	-10.02, 84.30	0.121
Effort (E)	38.91	14.23, 63.59	0.003
Effort ² (E ²)	-64.89	-90.90, -38.89	<0.001
Magnitude (M)	-54.88	-76.17, -33.59	<0.001
R:E	-41.49	-65.66, -17.31	0.001
R:E ²	11.25	-17.69, 40.19	0.446
R:M	44.86	7.00, 82.72	0.020
E:M	33.23	21.14, 45.32	<0.001
E ² :M	15.14	0.66, 29.62	0.040
R:E:M	16.46	-7.72, 40.64	0.182
R:E ² :M	-4.42	-33.38, 24.53	0.765
Observations	5945		

Notes: The final model was specificized as: Decision times ~ Recipient * Effort * Magnitude + Recipient * Effort² * Magnitude + (Recipient + Effort + Effort² + Magnitude | Participant). Both effort and magnitude levels were standardized before being entered into the model. Statistically significant *p* values (< 0.05, two-sided) are shown in bold. CI = confidence interval.

Table S5. Results of a mixed-effects logistic regression model predicting decision choices in the prosocial decision-making task.

Predictors	<i>b</i>	95% CI	<i>p</i>
Intercept	2.29	1.53, 3.04	<0.001
Recipient (R)	-1.74	-2.23, -1.14	<0.001
Effort (E)	-2.54	-2.93, -2.14	<0.001
Magnitude (M)	2.11	1.64, 2.59	<0.001
R:E	-0.18	-0.46, 0.10	0.216
R:M	-0.56	-0.82, -0.31	<0.001
E:M	0.15	0.02, 0.29	0.028
R:E:M	-0.12	-0.36, 0.11	0.303
Observations	5945		

Notes: The final model was specificized as: Decision choices ~ Recipient * Effort * Magnitude + (Recipient + Effort + Magnitude | Participant). Both effort and magnitude levels were standardized before being entered into the model. The *b* values are expressed in log-odds units. Statistically significant *p* values (< 0.05, two-sided) are shown in bold. CI = confidence interval.

Predictor	logK value			High-effort choice proportion		
	<i>b</i>	95% CI	<i>p</i>	<i>b</i>	95% CI	<i>p</i>
Intercept	2.88	1.86, 3.90	<0.001	2.83	1.81, 3.86	<0.001
Recipient (R)	-0.45	-1.09, 0.19	0.166	-0.42	-1.04, 0.20	0.180
Effort (E)	0.00	-0.27, 0.27	0.990	0.00	-0.26, 0.27	0.975
Magnitude (M)	0.42	0.13, 0.72	0.006	0.40	0.12, 0.69	0.007
Valence (V)	-1.16	-1.60, -0.71	<0.001	-1.10	-1.54, -0.67	<0.001
logK (K)/Choice (C)	-0.35	-0.85, 0.15	0.163	0.36	-0.14, 0.87	0.153
R:E	-0.72	-1.18, -0.26	0.002	-0.73	-1.18, -0.27	0.002
R:M	-0.41	-0.87, -0.04	0.077	-0.32	-0.77, 0.13	0.160
E:M	-0.16	-0.38, 0.07	0.177	-0.15	-0.37, 0.08	0.197
R:V	0.20	-0.68, 1.09	0.655	0.45	-0.42, 1.32	0.307
E:V	-0.37	-0.82, 0.08	0.106	-0.34	-0.78, 0.10	0.134
M:V	0.18	-0.26, 0.63	0.418	0.20	-0.23, 0.64	0.361
R:K/C	0.41	-0.21, 1.03	0.185	-0.60	-1.22, 0.02	0.057
E:K/C	0.19	-0.07, 0.45	0.144	-0.23	-0.49, 0.03	0.077
M:K/C	0.02	-0.25, 0.29	0.875	0.08	-0.19, 0.34	0.573
V:K/C	0.44	0.00, 0.89	0.050	-0.21	-0.65, 0.24	0.363
R:E:M	-0.39	-0.84, 0.06	0.093	-0.41	-0.85, 0.04	0.072
R:E:V	0.47	-0.43, 1.37	0.307	0.53	-0.35, 1.42	0.240
R:M:V	1.26	0.37, 2.14	0.005	1.13	0.26, 2.00	0.011
E:M:V	-0.09	-0.54, 0.37	0.710	0.01	-0.44, 0.45	0.979
R:E:K/C	-0.27	-0.74, 0.20	0.262	0.34	-0.12, 0.81	0.149
R:M:K/C	0.01	-0.46, 0.48	0.973	-0.13	-0.60, 0.34	0.594
E:M:K/C	-0.08	-0.31, 0.14	0.471	0.10	-0.13, 0.33	0.380
R:V:K/C	0.10	-0.79, 0.99	0.820	-0.14	-1.04, 0.75	0.752
E:V:K/C	0.34	-0.11, 0.80	0.139	-0.25	-0.71, 0.20	0.272
M:V:K/C	-0.43	-0.87, 0.02	0.060	0.38	-0.07, 0.83	0.094
R:E:M:V	0.43	-0.47, 1.33	0.350	0.58	-0.31, 1.46	0.203
R:E:M:K/C	0.58	0.12, 1.03	0.013	-0.58	-1.04, -0.12	0.014
R:E:V:K/C	0.82	-0.09, 1.72	0.077	-0.86	-1.78, 0.05	0.063
R:M:V:K/C	-0.66	-1.56, 0.23	0.145	0.44	-0.46, 1.33	0.341
E:M:V:K/C	0.06	-0.40, 0.52	0.794	-0.13	-0.59, 0.33	0.581
R:E:M:V:K/C	0.71	-0.20, 1.62	0.128	-0.39	-1.31, 0.53	0.406
Observations	7375			7566		

Notes. The final model was specified as: Amplitude ~ Effort * Magnitude * Valence * logK/Choice + (Recipient + Effort + Magnitude | Participant). Higher logK values are associated with lower high-effort choice proportions.

Table S6. Results of RewP models with discounting rate (logK) and high-effort choice proportions from fixed predictors.

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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors test the hypotheses, using an effort-exertion and an effort-based decision-making task, while recording brain dynamics with EEG, that the brain processes reward outcomes for effort differentially when they earned for themselves versus others.

Strengths:

The strengths of this experiment include what appears to be a novel finding of opposite signed effects of effort on the processing of reward outcomes when the recipient is self versus others. Also, the experiment is well-designed, the study seems sufficiently powered, and the data and code are publicly available.

Weaknesses:

There is some concern about the fact that participants report feeling less subjective effort, but also more disliking of tasks when they were earning rewards for others versus self. The concern is that participants worked with less vigor during self-versus-others trials and this may partly account for a key two-way Recipient x Effort interaction on the size of the Reward Positivity EEG component. Of note, participants took longer to complete tasks when working for others. While it is true that, in all cases, participants met the requisite task demands (they pressed the required number of buttons) they did so more sluggishly when earning rewards for others. The Authors argue that this reflects less motivation when working for others, which is a plausible explanation. The Authors also try to rule out this diminished vigor as a confounding explanation by showing that the two way interaction remains even when including reaction times (and also self-reported task liking) as a covariate. Nevertheless, it is possible that covariates do not fully account for the effects of differential motivation levels which would otherwise explain the two-way interaction. As such, I think a caveat is warranted regarding this particular result.

<https://doi.org/10.7554/eLife.103566.2.sa2>

Reviewer #2 (Public review):

Summary:

Measurements of the reward positivity, an electrophysiological component elicited during reward evaluation, have previously been used to understand how self-benefitting effort expenditure influences processing of rewards. The present study is the first to complement those measurements with electrophysiological reward after-effects of effort expenditure during prosocial acts. The results provide solid evidence that effort adds reward value when the recipient of the reward is the self but discounts reward value when the beneficiary is another individual.

Strengths:

An important strength of the study is that amount of effort, the prospective reward, the recipient of the reward, and whether the reward was actually gained or not were parametrically and orthogonally varied. In addition, the researchers examined whether the pattern of results generalized to decisions about future efforts. The sample size (N=40) and mixed-effects regression models are also appropriate for addressing the key research questions. Those conclusions are plausible and adequately supported by statistical analyses.

<https://doi.org/10.7554/eLife.103566.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

The authors test the hypotheses, using an effort-exertion and an effort-based decision-making task, while recording brain dynamics with EEG, that the brain processes reward outcomes for effort differentially when they earned for themselves versus others.

The strengths of this experiment include what appears to be a novel finding of opposite signed effects of effort on the processing of reward outcomes when the recipient is self versus others. Also, the experiment is well-designed, the study seems sufficiently powered, and the data and code are publicly available.

We thank Reviewer #1 for the affirmative appraisal of our manuscript as well as the thoughtful and insightful comments, which have enabled us to significantly improve the manuscript.

(1) Inferences rely heavily on the results of mixed effects models which may or may not be properly specified and are not supported by complementary analyses.

We thank Reviewer #1 for raising this critical issue of model specification. We have re-fitted our mixed-effects models and performed complementary analyses to validate the robustness of our findings. Specifically, we adopted the maximal converging random-effects structure (including random slopes for Recipient, Effort, and Magnitude where feasible) while ensuring model stability (see Responses to Reviewer #1's Recommendations point 2). Crucially, our primary findings, including the Recipient $\times$ Effort and Recipient $\times$ Effort $\times$ Magnitude interactions, remained robust. Furthermore, additional analyses confirmed that these results

were not confounded by factors such as response speed and subjective effort rating (see Responses to Reviewer #1's Recommendations point 5).

(2) Also, not all results hang together in a sensible way. For example, participants report feeling less subjective effort, but also more disliking of tasks when they were earning rewards for others versus self. Given that participants took longer to complete tasks when earning effort for others, it is conceivable that participants might have been working less hard for others versus themselves, and this may complicate the interpretation of results.

We thank Reviewer #1 for this insightful point (which also relates to Reviewer #3's point 5). In our study, participants were asked to rate three specific dimensions: Effort ("How much effort did you exert to complete each effort condition when earning rewards for yourself [or the other person]?"), Difficulty ("How much difficulty did you perceive in each effort condition when earning rewards for yourself [or the other person]?"), and liking ("How much did you like each effort condition when earning rewards for yourself [or the other person]?").

We acknowledge the Reviewer #1's concern that the lower subjective effort ratings for others seems contradictory to the higher disliking and longer completion times. We propose that in this paradigm, subjective effort ratings are susceptible to demand characteristics and likely captured motivational engagement (e.g., "how hard I tried" or "how willing I was") rather than perceived task demands. To disentangle these factors, we included a measure of perceived task difficulty, which is anchored in task properties and is less prone to social desirability biases (Harmon-Jones et al., 2020; Wright et al., 1990). We found no differences in perceived difficulty between self- and other-benefiting trials (Figure 2D), suggesting that the task demands were perceived as equivalent across conditions. To examine this interpretation more directly, we analyzed correlations among participants' ratings of difficulty, effort, and liking. As illustrated in Figure S1, we found no correlation between difficulty and effort ratings. Crucially, liking ratings were negatively correlated with difficulty ratings.

More importantly, our performance data contradict the interpretation that participants "worked less hard" for others in terms of task completion. While participants took longer to complete tasks for others, they maintained comparable, near-ceiling success rates for self (97%) and other (96%) recipients ($b = -0.46$, $p = 0.632$; Supplementary Table S1). This dissociation suggests that although participants were less motivated (e.g., lower subjective ratings, longer completion times, and greater disliking) to work for others, they ultimately exerted the necessary physical effort to achieve successful outcomes. Thus, the results consistently point to a decrease in prosocial motivation (consistent with prosocial apathy) rather than a failure of effort exertion.

Wright, R. A., Shaw, L. L., & Jones, C. R. (1990). Task demand and cardiovascular response magnitude: Further evidence of the mediating role of success importance. *Journal of Personality and Social Psychology*, 59(6), 1250-1260. <https://doi.org/10.1037/0022-3514.59.6.1250>

Harmon-Jones, E., Willoughby, C., Paul, K., & Harmon-Jones, C. (2020). The effect of perceived effort and perceived control on reward valuation: Using the reward positivity to test a dissonance theory prediction. *Biological Psychology*, 107910. <https://doi.org/10.1016/j.biopsycho.2020.107910>

Reviewer #2 (Public review):

Measurements of the reward positivity, an electrophysiological component elicited during reward evaluation, have previously been used to understand how self-benefitting effort expenditure influences the processing of rewards. The present study is the first to complement those measurements with electrophysiological reward after-effects of effort

expenditure during prosocial acts. The results provide solid evidence that effort adds reward value when the recipient of the reward is the self but discounts reward value when the beneficiary is another individual.

An important strength of the study is that the amount of effort, the prospective reward, the recipient of the reward, and whether the reward was actually gained or not were parametrically and orthogonally varied. In addition, the researchers examined whether the pattern of results generalized to decisions about future efforts. The sample size (N=40) and mixed-effects regression models are also appropriate for addressing the key research questions. Those conclusions are plausible and adequately supported by statistical analyses.

We appreciate Reviewer #2's positive appraisal of our manuscript. We are fortunate to receive your thoughtful and insightful suggestions and have revised the manuscript accordingly.

(1) Although the obtained results are highly plausible, I am concerned whether the reward positivity (RewP) and P3 were adequately measured. The RewP and P3 were defined as the average voltage values in the time intervals 300-400 ms and 300-440 ms after feedback onset, respectively. So they largely overlapped in time. Although the RewP measure was based on frontocentral electrodes (FC3, FCz, and FC4) and the P3 on posterior electrodes (P3, Pz, and P4), the scalp topographies in Figure 3 show that the RewP effects were larger at the posterior electrodes used for the P3 than at frontocentral electrodes. So there is a concern that the RewP and P3 were not independently measured. This type of problem can often be resolved using a spatiotemporal principal component analysis. My faith in the conclusions drawn would be further strengthened if the researchers extracted separate principal components for the RewP and P3 and performed their statistical analyses on the corresponding factor scores.

We thank Reviewer #2 for raising this issue. We would like to clarify that these two components were time-locked to different types of feedback and therefore reflect neural responses to distinct stages of the prosocial effort task. Specifically, the P3 was time-locked to performance feedback (the effort-completion cue; e.g., the tick shown in Figure 1B), whereas the RewP was time-locked to reward feedback (e.g., the display of “+0.6”). Thus, despite the numerical similarity in the post-stimulus windows, the components capture neural activity evoked by independent events separated in time, corresponding to the performance monitoring versus reward evaluation stages of the task. To avoid misunderstanding, we have made this distinction more explicit in the revised manuscript, which now reads, “Single-trial RewP amplitude was measured as mean voltage from 300 to 400 ms relative to reward feedback onset (i.e., reward delivery) over frontocentral channels (FC3, FCz, FC4). We also measured the parietal P3 (300–440 ms; averaged across P3, Pz, and P4) in response to performance feedback (i.e., effort completion), given its relationship with motivational salience (Bowyer et al., 2021; Ma et al., 2014)” (page 27, para. 1, lines 2–6).

Reviewer #3 (Public review):

This study investigates how effort influences reward evaluation during prosocial behaviour using EEG and experimental tasks manipulating effort and rewards for self and others. Results reveal a dissociable effect: for self-benefitting effort, rewards are evaluated more positively as effort increases, while for other-benefitting effort, rewards are evaluated less positively with higher effort. This dissociation, driven by reward system activation and independent of performance, provides new insights into the neural mechanisms of effort and reward in prosocial contexts.

This work makes a valuable contribution to the prosocial behaviour literature by addressing areas that previous research has largely overlooked. It highlights the

paradoxical effect of effort on reward evaluation and opens new avenues for investigating the mechanisms underlying this phenomenon. The study employs well-established tasks with robust replication in the literature and innovatively incorporates ERPs to examine effort-based prosocial decision-making - an area insufficiently explored in prior work. Moreover, the analyses are rigorous and grounded in established methodologies, further enhancing the study's credibility. These elements collectively underscore the study's significance in advancing our understanding of effort-based decision-making.

We thank Reviewer #3 for the positive assessment. We are particularly encouraged by the reviewer's recognition of our novel integration of ERPs to uncover the distinct effects of effort on reward evaluation for self versus others. We have carefully addressed the specific recommendations raised in the subsequent comments to further strengthen the rigor and clarity of the manuscript.

(1) Incomplete EEG Reporting: The methods indicate that EEG activity was recorded for both tasks; however, the manuscript reports EEG results only for the first task, omitting the decision-making task. If the authors claim a paradoxical effect of effort on self versus other rewards, as revealed by the RewP component, this should also be confirmed with results from the decision-making task. Omitting these findings weakens the overall argument.

We thank Reviewer #3 for giving us the opportunity to verify the specific roles of our two tasks. The primary aim of our study is to elucidate the neural after-effects of effort exertion on subsequent reward evaluation during prosocial acts. The prosocial effort task was specifically designed for this purpose, as it involves actual effort expenditure followed by reward outcomes. Furthermore, this task uses preset effort-reward combinations, ensuring balanced trial counts and adequate signal-to-noise ratios across conditions, a critical requirement for robust ERP analysis. In contrast, the prosocial decision-making task was included specifically to quantify behavioral preference (i.e., prosocial effort discounting) rather than neural reward processing. Specifically, this task involves choices without immediate effort execution and reward feedback, making it impossible to examine the neural after-effects of effort exertion. However, the decision-making task remains indispensable for our study structure: it provides an independent behavioral phenomenon of prosocial apathy, which allowed us to link individual differences in behavioral motivation to the neural dissociations observed in the prosocial effort tasks (as detailed in our Responses to Reviewer #3's 2). Thus, the two tasks provide complementary, rather than redundant, insights into the behavioral and neural mechanism of prosocial effort.

(2) Neural and Behavioural Integration: The neural results should be contrasted with behavioural data both within and between tasks. Specifically, the manuscript could examine whether neural responses predict performance within each task and whether neural and behavioural signals correlate across tasks. This integration would provide a more comprehensive understanding of the mechanisms at play.

We thank Reviewer #3 for this insightful and helpful suggestion. We agree that linking neural signatures with behavioral patterns is crucial for establishing the functional significance for our ERP findings. Regarding within-task association, it is important to note that the prosocial effort task was designed to require participants to exert fixed, preset levels of physical effort to earn uncertain rewards. This experimental control was necessary to standardize effort exertion across self-benefiting and other benefiting trials, thereby minimizing confounds such as differences in physical or perceived effort prior to the feedback phase. Indeed, the neural after-effects remained after controlling for these behavioral measures (i.e., response speed and self-reported effort; as detailed in responses to Reviewer #1's Recommendations point 5). Furthermore, unlike the prosocial effort task, the decision-making task inherently

precludes the examination of the neural after-effects of effort; therefore, within-task association in this task was not possible.

Given these considerations, we focused on the cross-task association. We examined whether the neural after-effects of effort (indexed by the RewP) in the prosocial effort task were modulated by individual differences in effort discounting. We used the K value estimated from the prosocial decision-making task as the index of effort discounting. We entered the K value (log-transformed and z-scored) as a continuous predictor into the mixed-effects models of RewP amplitudes. The full regression estimates for the model are presented in Table S1 (left).

We observed a significant four-way interaction among recipient, effort, magnitude, and K value ($b = 0.58$, $p = 0.013$). To decompose this complex interaction, we performed simple slopes analyses separately for self- and other-benefiting trials at high and low levels of reward magnitude and discounting rate (± 1 SD). As shown in Figure S2, for self-benefiting trials, the effort-enhancement effect on the RewP was significant only for participants with high discounting rates at low reward magnitude ($b = 1.02$, 95% CI = [0.22, 1.82], $p = 0.012$). In contrast, participants with low discounting rates exhibited no significant effort effect ($b = -0.37$, 95% CI = [-0.89, 0.15], $p = 0.159$). At high reward magnitude, simple slopes analyses detected no significant effort effects for either high ($b = 0.35$, 95% CI = [-0.44, 1.14], $p = 0.383$) or low ($b = 0.45$, 95% CI = [-0.07, 0.97], $p = 0.093$) discounting individuals. These findings strongly support the cognitive dissonance account (Aronson & Mills, 1959): those who find effort most aversive are most compelled to inflate the value of small rewards to justify their exertion. For these individuals, the completion of a costly action for a small reward may trigger a stronger internal justification effect, resulting in an amplified neural reward response.

For other-benefiting trials, participants with low discounting rates exhibited a significant effort-discounting effect at high reward magnitude ($b = -0.97$, 95% CI = [-1.74, -0.20], $p = 0.014$). In contrast, no significant effort effects were observed for participants with high discounting rates at either high ($b = -0.45$, 95% CI = [-0.97, 0.08], $p = 0.098$) or low ($b = -0.16$, 95% CI = [-0.69, 0.38], $p = 0.564$) reward magnitudes, nor for participants with low discounting rates at low reward magnitude ($b = 0.14$, 95% CI = [-0.64, 0.92], $p = 0.729$). These results suggest that the justification mechanism observed for self-benefiting effort appears absent for other-benefiting effort. Instead, we observed a persistent effort discounting before, during, and after effort expenditure, which was most pronounced in individuals with low effort sensitivity (low K) when reward magnitude was high. This seemingly paradoxical pattern might be interpreted through the lens of disadvantageous inequity aversion (Fehr & Schmidt, 1999). Specifically, the combination of high personal effort and high monetary reward for another person creates a salient disparity between the participant's incurred cost and the recipient's gain. Although low- K individuals are behaviorally willing to tolerate this cost, their neural valuation system may nonetheless track the "unfairness" of this asymmetry, thereby attenuating the neural reward signal (Tricomi et al., 2010). These insights suggest that facilitating prosocial behavior may require not just lowering costs, but potentially framing outcomes to trigger the effort justification mechanisms that drive the effort paradox observed in self-benefiting acts (Inzlicht & Campbell, 2022).

To confirm this four-way interaction, we also replaced the high-effort choice proportions in the decision-making task and observed a similar four-way interaction among recipient, effort, magnitude, and high-effort choice proportions ($b = -0.58$, $p = 0.014$; see Table S1 for detailed regression estimates). Together, this cross-task analysis not only provides a more comprehensive understanding of the mechanisms at play but also justifies the inclusion of the prosocial decision-making task. We sincerely thank Reviewer #3' for this valuable suggestion, which has significantly strengthened our manuscript. We have included this

analysis (page 16, para. 2; page 17, paras. 1–2) and discussed the results (page 20, para. 2, lines 10–15; page 20, para. 3; page 21, para. 1, lines 1–8) in the revised manuscript.

Aronson, E., & Mills, J. (1959). The effect of severity of initiation on liking for a group. *The Journal of Abnormal and Social Psychology*, 59(2), 177–181. <https://doi.org/10.1037/h0047195>

Fehr, E., & Schmidt, K. M. (1999). A theory of fairness, competition, and cooperation. *The Quarterly Journal of Economics*, 114(3), 817–868. <http://www.jstor.org/stable/2586885>

Tricomi, E., Rangel, A., Camerer, C. F., & O'Doherty, J. P. (2010). Neural evidence for inequality-averse social preferences. *Nature*, 463(7284), 1089–1091. <https://doi.org/10.1038/nature08785>

(3) Success Rate and Model Structure: The manuscript does not clearly report the success rate in the prosocial effort task. If success rates are low, risk aversion could confound the results. Additionally, it is unclear whether the models accounted for successful versus unsuccessful trials or whether success was included as a covariate. If this information is present, it needs to be explicitly clarified. The exclusion criteria for unsuccessful trials in both tasks should also be detailed. Moreover, the decision to exclude electrodes as independent variables in the models warrants an explanation.

We appreciate the opportunity to clarify these points. In the revised manuscript, we have now explicitly reported the descriptive statistics and the results of a mixed-effects logistic model on response success in the revised manuscript (page 8, para. 1, lines 2–4; Supplementary Table S1). Participants achieved similarly high success rates in both self ($M = 97\%$) and other trials ($M = 96\%$; Figure S3). As shown in Table S2, success rates decreased as effort increased ($b = -4.77$, $p < 0.001$). However, no other effects reached significance ($ps > 0.245$). These near-ceiling success rates indicate strong task engagement and effectively rule out risk aversion as a potential confound.

Regarding model structure, we excluded unsuccessful trials from statistical analyses because they were rare and distributed equally across conditions. Given the near-ceiling performance, we did not include success rate as a covariate, as it offers limited variance.

Finally, we did not include electrodes as an independent variable because our hypotheses focused on condition effects rather than topographic differences. Following established research (e.g., Krigolson, 2018; Proudfit, 2015), we averaged RewP amplitudes across a frontocentral cluster (FC3, FCz, and FC4) and P3 amplitudes across a parietal cluster (P3, Pz, and P4), where activity is typically maximal. Averaging across these theoretically grounded clusters improves the signal-to-noise ratio and provides more reliable estimates of the underlying components. We have explicitly included this rationale in the revised manuscript, which reads, “Data were averaged across the selected electrode clusters to improve signal-to-noise ratio and reliability” (page 27, para. 1, lines 9–10).

Proudfit, G. H. (2015). The reward positivity: From basic research on reward to a biomarker for depression. *Psychophysiology*, 52(4), 449–459. <https://doi.org/10.1111/psyp.12370>

Krigolson, O. E. (2018). Event-related brain potentials and the study of reward processing: Methodological considerations. *Int J Psychophysiol*, 132(Pt B), 175–183. <https://doi.org/10.1016/j.ijpsycho.2017.11.007>

(4) Prosocial Decision Computational Modelling: The prosocial decision task largely replicates prior behavioural findings but misses the opportunity to directly test the hypotheses derived from neural data in the prosocial effort task. If the authors propose a paradoxical effect of effort on self-rewards and an inverse effect for prosocial effort, this could be formalised in a computational model. A model comparison could evaluate the proposed mechanism against alternative theories, incorporating the complex interplay

of effort and reward for self and others. Furthermore, these parameters should be correlated with neural signals, adding a critical layer of evidence to the claims. As it is, the inclusion of the prosocial decision task seems irrelevant.

We thank Reviewer #3 for this thoughtful suggestion regarding the value of computational modelling. We fully agree that formalizing mechanisms is crucial, but we would like to clarify why a computational model of decision-making cannot directly capture the paradoxical after-effects observed in our neural data. The paradoxical after-effect of effort exertion we report refers to experienced utility (i.e., how prior costs modulate the hedonic consumption of a reward), whereas the decision task measures decision utility (i.e., how prospective costs and benefits are integrated to guide choice). We included the prosocial decision task to establish a behavioral baseline and replicate the well-documented phenomenon of prosocial apathy. Consistent with prior work (e.g., Lockwood et al., 2017; Lockwood et al., 2022), our data show that at the decision stage (ex-ante), effort functions as a universal cost: participants discounted rewards for both self and others, differing only quantitatively (steeper discounting for others). It is only after effort is exerted (ex-post) that the pattern reverses: effort is valued for self but remains costly for others, representing a qualitative shift. Crucially, incorporating a "paradoxical valuation" parameter (i.e., effort as a reward) into our decision model would mathematically contradict the behavioral reality. Since participants actively avoided high-effort options, a model assuming effort adds value might fail to fit the choice data. The theoretical novelty of our study lies precisely in this temporal dissociation: whereas self-benefiting effort paradoxically enhances reward valuation, other-benefiting effort induces a persistent reward devaluation.

To address the reviewer's interest in bridging these two domains, we examined whether these distinct stages are linked at the level of individual differences. We hypothesized that an individual's sensitivity to prospective effort cost (discounting rate K) might modulate their susceptibility to the retrospective neural after-effect. As detailed in our Responses to Reviewer #3's point 2, we found that for self-benefiting trials, high-discounting individuals showed an effort-enhancement effect on the RewP at low reward magnitude, while for other-benefiting trials, low-discounting individuals exhibited effort-discounting effects at high reward magnitude. We sincerely thank Reviewer #3' for this valuable suggestion, which has successfully correlated the two tasks and facilitated our understanding of the mechanisms at play.

Lockwood, P. L., Hamonet, M., Zhang, S. H., Ratnavel, A., Salmony, F. U., Husain, M., & Apps, M. A. J. (2017). Prosocial apathy for helping others when effort is required. *Nat Hum Behav*, 1(7), 0131. <https://doi.org/10.1038/s41562-017-0131>

Lockwood, P. L., Wittmann, M. K., Nili, H., Matsumoto-Ryan, M., Abdurahman, A., Cutler, J., Husain, M., & Apps, M. A. J. (2022). Distinct neural representations for prosocial and self-benefiting effort. *Curr Biol*, 32(19), 4172-4185 e4177. <https://doi.org/10.1016/j.cub.2022.08.010>

(5) Contradiction Between Effort Perception and Neural Results: Participants reported effort as less effortful in the prosocial condition compared to the self condition, which seems contradictory to the neural findings and the authors' interpretation. If effort has a discounting effect on rewards for others, one might expect it to feel more effortful. How do the authors reconcile these results? Additionally, the relationship between behavioural data and neural responses should be examined to clarify these inconsistencies.

This point aligns with the issues raised in Reviewer #1's point 2. We acknowledge the apparent discrepancy between lower reported effort in the prosocial condition and the neural discounting effect. As detailed in our Responses to Reviewer #1's point 2, we reconcile this by proposing that subjective effort ratings in this paradigm likely reflect motivational engagement (e.g., "how hard I tried" or "how willing I was") rather than perceived task

demands. Under this interpretation, the lower effort ratings for others reflect a withdrawal of engagement (consistent with prosocial apathy), which conceptually aligns with, rather than contradicts, the neural discounting effect. To validate this, we contrasted effort ratings with difficulty ratings (a more reliable index of objective demand). Our correlational analysis revealed no significant relationship between difficulty and effort ratings ($r = -0.21$, $p = 0.196$), suggesting that they capture distinct constructs. Furthermore, liking ratings were negatively correlated with difficulty ratings ($r = -0.43$, $p = 0.011$) but not with effort ratings ($r = 0.32$, $p = 0.061$), further dissociating the two measures. Crucially, as detailed in our Responses to Reviewer #1's Recommendations point 5, our RewP effects remained significant even after controlling for individual effort ratings. This demonstrates that the neural effort-discounting effect for others is a physiological signature that operates independently of the subjective report bias.

(6) Necessary Revisions to Manuscript: If the authors address the issues above, corresponding updates to the introduction and discussion sections could strengthen the narrative and align the manuscript with the additional analyses.

We thank Reviewer #3 for the above insightful and helpful comments. We have carefully addressed these issues raised above and have updated the manuscript accordingly, including abstract, introduction, result, and discussion sections.

Recommendations for the Authors:

Reviewer #1 (Recommendations for the authors):

Major comments:

(1) The two biggest concerns I have are

- Whether the mixed-effect models are properly specified, and*
- Whether the main interaction between the Recipient and effort on the reward positivity (RewP) reflects different levels of effort exertion when working for self versus others.*

We thank Reviewer #1 for identifying these two critical issues. We have carefully considered these points and conducted additional analyses to address them. Below, we provide a detailed response to each concern, explaining how we have improved the model specification and ruled out alternative interpretations regarding effort exertion.

(2) On the first point, I noticed that the authors selectively excluded random effects for Effort and Magnitude when regressing RewP on Effort, Magnitude, Recipient, and Valence. This is important because the key result in the paper is a fixed effect two-way interaction between Recipient and Effort and a three-way interaction between Recipient, Effort, and Magnitude. It is not clear that these results will remain significant when Effort and Magnitude are included as random effects in the model. Thus the authors should justify their exclusion as random effects, and/or show that the results don't depend on including those random effects in the model. The same logic applies to the specification of other mixed effects models (e.g. the effect of Magnitude in the model predicting RTs).

We thank Reviewer #1 for raising this important methodological point. We fully agree that including random slopes wherever possible reduces Type 1 error rates and yields more conservative tests of fixed effects. In our analyses, we determined the random effects structure for each model using singular value decomposition (SVD). Specifically, we began with a maximal model that included by-participant random slopes for all main effects and interactions as well as a participant-level random intercept. When the model failed to converge or yielded a singular fit, we applied SVD to identify redundant dimensions (i.e., components explaining zero variance) and iteratively removed these terms until convergence

was achieved. This procedure allowed us to retain the maximal converging random-effects structure while ensuring model stability. We have clarified this procedure in the revised manuscript as follows, “For each model, we fitted the maximal random-effects structure and, when the model was overparameterized, used singular value decomposition to simplify the random-effects structure until the model converged” (page 28, para. 1, lines 5–8).

Regarding the RewP model, including all variables (i.e., Recipient, Effort, Magnitude, and Valence) in the random-effects structure resulted in a boundary (singular) fit. Examination of the variance-covariance structure of the random effects revealed that the random slopes for Valence and Magnitude were perfectly negatively correlated ($r = -1.00$), indicating severe overparameterization. In our original submission, we removed the random slopes for Effort and Magnitude because the SVD analysis indicated redundant dimensions in the model structure.

However, we agree with the Reviewer that retaining slopes for variables involved in key interactions is crucial. Therefore, we re-evaluated the model strategy: instead of removing Effort and Magnitude, we removed the random slope for Valence (which was the primary source of the perfect correlation). This modification successfully resolved the singularity while allowing us to retain the random slopes for the critical variables (i.e., Effort and Magnitude).

Critically, this updated model yielded the same pattern of results as our original submission: the two-way interaction between Recipient and Effort and the three-way interaction between Recipient, Effort, and Magnitude remained significant (see Table S3). As expected, including the random slopes for Effort and Magnitude yielded a more conservative test of the fixed effects. While the critical three-way interaction remained significant ($p = 0.019$), the simple slope for the Self condition at high reward magnitude shifted slightly from significant ($p = 0.041$) to marginally significant ($p = 0.056$). However, the effect size remained largely unchanged ($b = 0.42$ vs. original $b = 0.43$), and the dissociation pattern, where self-benefiting trials show a positive trend while other-benefiting trials show a significant negative slope, remains robust and is statistically supported by the significant interaction. We have adopted this updated model in the revised manuscript and updated the relevant sections accordingly. Finally, note that we have removed the RewP table from the Supplementary Materials because the RewP model results are now presented as a figure in the main text (as suggested by Reviewer #1’s Recommendations point 3).

We have also carefully verified the random effects structures for other mixed-effects models, including the RT and Performance-P3 models in the prosocial effort task, as well as the decision time and decision choice models in the prosocial decision-making task. The updated information is detailed as follows:

Regarding the RT model, we replaced it with a more reasonable model of response speed (button presses per second), as suggested by Reviewer #1 (see our responses to Reviewer #1’s Recommendations point 4 for details).

Regarding the performance-P3 model, the random-effects structure could only support Effort, as in our original submission; thus, the results remain unchanged.

Regarding the decision time model, we have updated our results to include the quadratic effort term, as suggested by Reviewer #1 (see our responses to Reviewer #1’s Recommendations point 6 for details).

Regarding the decision choice model, we included Recipient, Effort, and Magnitude in the random-effects structure. As shown in Table S4, the results remain largely consistent with the original model, except for a newly significant interaction between effort and magnitude. Follow-up simple slopes analyses revealed that the discounted effect of effort was more

pronounced at low reward magnitude ($M - 1SD$: $b = -2.69$, 95% CI = $[-3.09, -2.29]$, $p < 0.001$) than at high reward magnitude ($M + 1SD$: $b = -2.38$, 95% CI = $[-2.82, -1.94]$, $p < 0.001$).

In summary, we have improved the model specification following Reviewer #1's suggestion. Crucially, the results remain qualitatively consistent with our original findings. We have updated the Results section, figures (Figures 2, 4, and 5), and OSF documents (including a new R Markdown file and an HTML output file detailing the final results) to reflect these analyses. Additionally, we have explicitly stated the method used for calculating p -values in the mixed-effects models (page 28, para. 1, lines 8–10), which was omitted in the original submission.

(3) Regarding the mixed models, it would also be good to show a graphical depiction summarizing key effects (e.g. the Recipient by Effort interaction on RewP) rather than just showing the predictions of the fitted mixed effects models.

This point is well-taken. Please see Figure S4, which visualizes the key effects and has now been included in the revised manuscript as Figure 4A.

(4) Finally, regarding the mixed effect models of RTs - given the common finding that RTs are not normally distributed, the Authors might be better off regressing 1/RT (interpreted as speed rather than latency) since 1/RT will often make distributions less asymmetric and heavy-tailed.

We thank Reviewer #1 for this helpful suggestion regarding data distribution. In our original analysis, the dependent variable was “completion time” (i.e., the latency to complete the required button presses with the 6-s window). We agree that these raw latency data exhibited characteristic non-normality (see Figure S5, Left). Based on Reviewer #1's suggestion, we adopted “response speed” (calculated as button presses per second) as the dependent variable. As expected, this transformation substantially improved the normality of the distribution (see Figure S5, Right). We have refitted the mixed-effects model using this speed metric. Critically, the results largely replicated the patterns observed in our original model, with the exception that the main effect of reward magnitude did not reach significance in the speed model (see Table 5). Given the superior distributional properties of the speed metric, we have replaced the original latency analysis with the response speed model in the revised manuscript. We have updated the Results section (page 8, para. 1, lines 4–9) and Figures 2B–C accordingly.

(5) Regarding the level of effort exerted, there are two reasons to suspect that participants exerted less for others versus themselves. The first is that they were slower to complete the button pressing for others versus themselves. The second is that they reported paradoxically less subjective effort for others versus self (paradoxical because they also reported liking the task less for others versus self). The explanation for both may be that they exerted less effort for others versus self and this has important implications for interpreting the main effects. If they exerted less effort for others, this may partly account for the key Recipient:Effort and Recipient:Effort:Magnitude interactions in the mixed effects regression of RewP. Do either median effort durations or self-reported effort predict the magnitude of the Recipient:Effort and Recipient:Effort:Magnitude interactions (if these were included as random effects)? If so, that would provide evidence supporting this story. Alternatively, if median durations or self-reported effort were included as covariates, do these interactions still obtain? In any case, the Authors should include caveats regarding this potential explanation of the self-versus-other interactions with effort and magnitude on the RewP" (or explain why this can not explain the interactions).

We thank Reviewer #1 for raising this important interpretational issue. We acknowledge the concern that differences in physical exertion or perceived effort could potentially confound

the neural findings. However, we argue that the observed RewP effects are not driven by these factors for several reasons.

First, the prosocial effort task enforced fixed effort thresholds (10%–90% of their maximum effort level) across self-benefiting and other-benefiting trials. Importantly, participants achieved ceiling-level success rates that were highly comparable between self-benefiting (97%) and other-benefiting (96%) trials, indicating that they successfully exerted the required effort across conditions.

Second, regarding the slower response speed for others (we used response speed instead of completion time, as the former is more suitable for statistical analysis; see details in Responses to Reviewer #1's Recommendations point 4), we interpret this as a reduction in motivation rather than a reduction in the amount of effort exerted. Similarly, as detailed in our Responses to Reviewer #1's point 2, subjective effort ratings in this paradigm appear to be influenced by demand characteristics and do not reliably track physical exertion. For instance, liking ratings were associated with difficulty ($r = -0.43$, $p = 0.011$) instead of effort ($r = 0.32$, $p = 0.061$) ratings.

To empirically rule out the possibility that these behavioral differences account for the neural effect, we followed the reviewer's suggestion and re-ran the mixed-effects model predicting RewP amplitudes with trial-by-trial response speed and subjective effort rating included as covariates. These control analyses revealed that neither response speed ($b = -0.07$, $p = 0.614$) nor self-reported effort ($b = 0.10$, $p = 0.186$) significantly predicted RewP amplitudes (see Table S6). Most importantly, the key interactions of interest (Recipient $\times$ Effort and Recipient $\times$ Effort $\times$ Magnitude) remained significant and virtually unchanged. These findings suggest that the observed neural after-effects of prosocial effort are not driven by variations in motor execution or perceived effort.

Minor comments:

(6) In Figure 5A a quadratic effect (not a linear effect) seems fairly obvious in decision times as a function of effort level. This makes sense given that participants are close to indifference, on average, around the 50-70% effort level. I recommend fitting a model that has a quadratic predictor and not just a linear predictor when regression decision times on effort levels.

We thank Reviewer #1 for this insightful suggestion. We agree that decision times likely track decision conflict, which typically peaks near indifference points (e.g., moderate effort levels). Accordingly, we reanalyzed the decision time data using a mixed-effects model that included both linear and quadratic terms for effort. As detailed in Table S7, this analysis revealed a significant quadratic main effect of effort, which was further qualified by a significant interaction between the quadratic effort term and reward magnitude. Decomposition of this interaction (Figure S6) revealed that the quadratic effort effect was more pronounced at low reward magnitude ($M - 1SD$: $b = -160.10$, 95% CI = $[-218.30, -101.90]$, $p < 0.001$) than at high reward magnitude ($M + 1SD$: $b = -99.50$, 95% CI = $[-157.60, -41.40]$, $p = 0.001$). However, we found no significant interactions involving the quadratic effort term and recipient. We have updated the Results section (page 13, para. 2; page 14, para. 1) and Figures 5A–B (right panel) to reflect these findings.

(7) The distinction between the effort and decision-making tasks wasn't super clear from the main text. A sentence early on in the results section could be useful for readers' understanding.

This point is well taken. In the revised manuscript, we have clarified this distinction at the beginning of the Results section (page 6, para. 2, lines 1–10). In addition, we have explicitly indicated the corresponding task within each subsection heading in the Results:

“2.1 Investing effort for others is less motivating than for self in the prosocial effort task” (page 7)

“2.2 Effort adds reward value for self but discounts reward value for others in the prosocial effort task” (page 9)

“2.3 Reward is devalued by effort to a higher degree for others than for self in the prosocial decision-making task” (page 13)

| (8) To what does “three trials” refer to on lines 143-144?

Thank you for raising this point. Participants completed three trials in which they were asked to press a button as rapidly as possible with their non-dominant pinky finger for 6000 ms. The maximum effort level was operationalized as the average button-press count across the three trials. To improve clarity, we have also provided more detailed description in the Results section, which reads: “The mean maximum effort level (i.e., the average button-press count across three 6000-ms trials; see Procedure for details)” (page 7, para. 1, lines 1–2).

| (9) It is unclear how the authors select their time windows for ERP analyses.

We thank Reviewer #1 for this comment. Measurement parameters (i.e., time windows and channel sites) were determined based on the grand-averaged ERP waveforms and topographic maps collapsed across all conditions. This procedure is orthogonal to the conditions of interest and prevents bias in the selection of measurement windows and channels, consistent with the “orthogonal selection approach” (Luck & Gaspelin, 2017). We have clarified this point in the revised manuscript, which now reads, “Measurement parameters (time windows and channel sites) were determined from the grand-averaged ERP waveforms and topographic maps collapsed across all conditions, which was thus orthogonal to the conditions of interest (Luck & Gaspelin, 2017)” (page 27, para. 1, lines 6–9).

Luck, S., & Gaspelin, N. (2017). How to get statistically significant effects in any ERP experiment (and why you shouldn't). *Psychophysiology*, 54(1), 146-157.

| (10) There are a few typos throughout. For example, Line 124 should read “other half benefitted...”, Line 127 should read “interest at each effort level...”, “following” on Line 369, and Supplemental table titles incorrectly spell the word “Results”.

We thank Reviewer #1 for catching these errors. We have corrected all the specific typos noted (page 6, para. 2, lines 11 and 15; page 22, para. 3, line 2; Supplementary Table S2). Furthermore, we have conducted a thorough proofreading of the entire text and supplementary materials to ensure linguistic accuracy and consistency throughout the manuscript.

Reviewer #2 (Recommendations for the authors):

Minor comments:

(1) Lines 84-86. “The RewP ... has its neural sources in the anterior cingulate cortex (Gehring & Willoughby, 2002) and ventral striatum (Foti et al., 2011).” This is a better reference for the ACC source: <https://pubmed.ncbi.nlm.nih.gov/23973408/>. And perhaps remove the reference to the ventral striatum; most people would agree that activity in the ventral striatum cannot be measured with scalp EEG.

We thank Reviewer #2 for providing the updated reference, which has been cited in the revised manuscript. We agree that activity in the VS cannot be reliably measured with scalp EEG and thus have removed the reference to the VS. The revised sentence now reads, “... has

its neural sources in the anterior cingulate cortex (Gehring & Willoughby, 2002; Hauser et al., 2014)” (page 4, para. 2, lines 12–13).

(2) Lines 152-153. What exactly is shown in Figure 2A? How did the authors average across subjects?

We thank Reviewer #2 for raising this issue. Figure 2A depicts the distribution of the maximum effort level, defined as the average button-press count across three 6000-ms trials completed before the prosocial effort task. In these trials, participants were instructed to press the button as rapidly as possible with their non-dominant pinky fingers. To improve clarity, we have revised the figure caption as: “(A) Distribution of the maximum effort level (i.e., the average button-press count across three 6000-ms trials) across participants” (Figure 2).

(3) Lines 160-164. “As expected (Figure 2D), participants perceived increased effort as more difficult ... and more disliking ($b = -0.62$, $p < 0.001$) when the beneficiary was others than themselves.” Does this sentence describe the main effect of the beneficiary or the interaction between beneficiary and effort level, as the start of the sentence (“increased effort”) suggests?

We thank Reviewer #2 for pointing out this ambiguity. The sentence describes the main effect of beneficiary rather than the interaction between beneficiary and effort level. In the revised manuscript, we have rephrased the sentence as: “They felt less effort ($b = -0.32$, $p = 0.019$) and more disliking ($b = -0.62$, $p = 0.001$) for other-benefiting trials compared to self-benefiting trials” (page 9, para. 1, lines 4–6).

(4) Lines 195-196. “..., we conducted post-hoc simple slopes analyses at -1 SD (“Low”) and + SD (“High”) reward magnitude.” I did not understand what the authors meant with these reward magnitudes, given that the actual potential rewards were ¥0.2, ¥0.4, ¥0.6, ¥0.8, and ¥1.0.

In our analyses, the actual reward magnitudes (¥0.2, ¥0.4, ¥0.6, ¥0.8, and ¥1.0) were z-scored and entered as a continuous regressor in the mixed-effects models. Post-hoc simple slopes analyses were then conducted at ± 1 SD from the mean of the z-scored reward magnitude. To clarify, we have revised the sentence as “... we conducted post-hoc simple slopes analyses at 1 standard deviation (SD) below (“Low”) and above (“High”) the mean reward magnitude” (page 11, para. 2, lines 8–9). This standard method for testing simple effects for continuous predictors is recommended by Aiken and West (1991). Aiken, L. S., West, S. G., & Reno, R. R. (1991). Multiple regression: Testing and interpreting interactions. Sage.

(5) Lines 253 and 275. I would not call this a computational model. The authors fit a curve to data, there is no model of the computations involved.

This point is well taken. We have replaced “computational model” with “discounting” (Figure 5) and “parabolic discounting model” (page 15, para. 1, line 15).

(6) Line 710. Figure S1 does not show topographic maps of the P3, as the figure caption suggests.

We thank Reviewer #2 for identifying this oversight. We have now included topographic maps of the P3 in Figure S1.

(7) Please check language in lines 33 (effect between), 38 (shape), 49 (highest cost form?), 74 (tunning), 90 (omit following), 127 (interest on at each effort level), 135 (press buttons >> rapidly press a button?), 142 (motivated), 219 (should low be high?), 265-266 (missing word), 275 (confirmed by following), 292 (an action can be effortful, a feeling cannot),

315 (*when it comes into*), 330-331 (*data is plural; the aftereffect of prosocial effect*), 387 (*interest on at each effort level*), 405 (*should quickly be often?*).

We thank Reviewer #2 for the careful review and feedback about these language issues. We have revised all the phrasing you identified. The corrections are as follows:

Line 33: “effect between” has been changed to “effects for” (page 2, para. 1, line 6).

Line 38: “shape” has been updated to “shapes” (page 2, para. 1, line 13).

Line 49: “highest cost form?” has been revised to “the most common cost type” (page 3, para. 1, lines 7–8).

Line 74: “tunning” has been corrected to “tuning” (page 4, para. 2, line 1).

Line 90: omit following. Done (page 5, para. 1, line 2).

Line 127: “interest on at each effort level” has been corrected to “liking for each effort level” (page 6, para. 2, line 15).

Line 135: “press buttons” has been updated to “rapidly press a button” (the caption of Figure 1).

Line 142: “motivated” has been revised to “motivating” (page 7).

Line 219: should low be high? Yes, we have corrected this (the caption of Figure 4).

Lines 265–266: The missing word “with” has been inserted (page 15, para. 1, line 2).

Line 275: “confirmed by following” has been revised as “corroborated by a parabolic ...” (page 15, para. 1, line 15).

Line 292: an action can be effortful, a feeling cannot. We have changed the word “effortful” to “effort” (page 18, para. 2, line 3).

Line 315: “when it comes into” has been revised to “when it came to” (page 19, para. 1, line 10).

Lines 330–331: These two expressions have been revised to “our data establish ...” and “the after-effect of prosocial effort” (page 20, para. 1, lines 2–3).

Line 387: “interest on at each effort level” has been corrected to “interest at each effort level” (page 23, para. 2, line 5).

Line 405: should quickly be often? We agree that “quickly” might imply latency or speed of a single press, whereas the task required maximizing the frequency of presses within the time window. To capture this meaning accurately, we have revised the phrase to “pressed a button as rapidly as possible” (implying repetition rate) in the revised manuscript (page 24, para. 2, lines 3–4).

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